ABSTRACT ALGEBRA

Complete Theory, Rigorous Proofs, and Problem Sets

Adapted from *Abstract Algebra* (3rd Edition) by David S. Dummit and Richard M. Foote

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Course Coverage and Syllabus Structure:

- **Unit I:** Homomorphisms, Isomorphisms, Subgroups, Center, Subgroups Generated by Subsets, Cosets, Lagrange's Theorem, Euler's, Fermat's, and Wilson's Theorems.
- **Unit II:** Products of Groups, Normal Subgroups, Quotient Groups, The Four Isomorphism Theorems, Commutator Subgroups and Abelianization.
- **Unit III:** Symmetric Groups, Cycle Decompositions, Transpositions, Even/Odd Permutations, Alternating Groups, Normal Subgroups of S_n, Simplicity of A_n.
- **Unit IV:** Group Actions, Orbits, Stabilizers (Isotropy), Regular and Conjugation Actions, Class Equation, Groups of Order p^2, Rings, Subrings, Gaussian Integers, Matrix Rings, Quaternions, Ideals, and Quotient Rings.
- **Unit V:** Integral Domains, Division Rings, Fields, Field of Fractions, Polynomial Rings over Fields, Division Algorithm, Ideals of $F[x]$, Algebras over $\mathbb{R}$ $(\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O})$, Frobenius and Hurwitz Classification Theorems.

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Contents

Chapter 1

Group Foundations, Subgroups, and Lagrange's Theorem

This chapter establishes the axiomatic foundation of modern group theory: homomorphisms, isomorphisms, subgroups, subgroup generation, centralizers, normalizers, the center of a group, coset partitions, and Lagrange's Theorem alongside its number-theoretic consequences (Euler's, Fermat's, and Wilson's Theorems). Every result is accompanied by an exhaustive, step-by-step proof with all algebraic justifications explicitly detailed.

1.1 Homomorphisms and Isomorphisms

In this section, we study structure-preserving maps between groups.

Definition 1.1 (Group Homomorphism)**.** Let $(G, \star)$ and $(H, \circ)$ be two groups. A mapping $\varphi : G \to H$ is called a **group homomorphism** (or simply a **homomorphism**) if for all $x, y \in G$:

$$\varphi(x \star y) = \varphi(x) \circ \varphi(y). \tag{1.1}$$

When the binary operations are understood from context, this is written in standard multiplicative notation as:

$$\varphi(xy) = \varphi(x)\varphi(y). \tag{1.2}$$

Proposition 1.2 (Basic Structural Properties of Homomorphisms)**.** *Let $\varphi : G \to H$ be a group homomorphism. Then:*

(i) *$\varphi(1_G) = 1_H$, where 1_G and 1_H denote the identity elements of G and H, respectively.*

(ii) *$\varphi(g^{-1}) = \big(\varphi(g)\big)^{-1}$ for all $g \in G$.*

(iii) *$\varphi(g^n) = \big(\varphi(g)\big)^n$ for all $n \in \mathbb{Z}$.*

(iv) *If $|g| = k < \infty$, then $|\varphi(g)|$ divides k.*

Proof. **(i) Preservation of Identity Element:** By the definition of the identity element in G, we have:

$$1_G \cdot 1_G = 1_G.$$

Applying the homomorphism property of φ to both sides yields:

$$\varphi(1_G)\varphi(1_G) = \varphi(1_G \cdot 1_G) = \varphi(1_G).$$

Since $\varphi(1_G) \in H$, it possesses a unique two-sided inverse $\big(\varphi(1_G)\big)^{-1}$ in H. Multiplying both sides of the equation on the right by this inverse:

$$\begin{aligned}
\big(\varphi(1_G)\varphi(1_G)\big)\big(\varphi(1_G)\big)^{-1} &= \varphi(1_G)\big(\varphi(1_G)\big)^{-1} \\
\varphi(1_G)\Big(\varphi(1_G)\big(\varphi(1_G)\big)^{-1}\Big) &= 1_H \quad \text{(by associativity in } H\text{)} \\
\varphi(1_G) \cdot 1_H &= 1_H \\
\varphi(1_G) &= 1_H.
\end{aligned}$$

(ii) Preservation of Inverses: Let $g \in G$. By definition of the group inverse, $g \cdot g^{-1} = 1_G$. Applying φ:

$$\begin{aligned} \varphi(g)\varphi(g^{-1}) &= \varphi(g \cdot g^{-1}) \quad \text{(since } \varphi \text{ is a homomorphism)} \\ &= \varphi(1_G) \\ &= 1_H \quad \text{(by part (i))}. \end{aligned}$$

Similarly, $g^{-1} \cdot g = 1_G$ implies $\varphi(g^{-1})\varphi(g) = 1_H$. By the uniqueness of the inverse element in the group H, it follows that:

$$\varphi(g^{-1}) = \big(\varphi(g)\big)^{-1}.$$

(iii) Preservation of Integer Powers: We establish this for all $n \in \mathbb{Z}$ in three cases:

- *Case $n > 0$ (Mathematical Induction):* For $n = 1$, $\varphi(g^1) = \varphi(g) = \big(\varphi(g)\big)^1$, so the base case holds. Assume inductively that $\varphi(g^k) = \big(\varphi(g)\big)^k$ for some positive integer $k \geq 1$. Then for $n = k + 1$:

$$\begin{aligned} \varphi(g^{k+1}) &= \varphi(g \cdot g^k) \\ &= \varphi(g)\varphi(g^k) \quad \text{(homomorphism property)} \\ &= \varphi(g)\big(\varphi(g)\big)^k \quad \text{(induction hypothesis)} \\ &= \big(\varphi(g)\big)^{k+1}. \end{aligned}$$

  Thus the statement holds for all $n \in \mathbb{Z}^+$.

- *Case $n = 0$:* By definition of zeroth power, $g^0 = 1_G$. Thus:

$$\varphi(g^0) = \varphi(1_G) = 1_H = \big(\varphi(g)\big)^0.$$

- *Case $n < 0$:* Let $n = -m$ where $m > 0$. Using the definition of negative powers and parts (i)–(ii):

$$\begin{aligned} \varphi(g^n) &= \varphi(g^{-m}) = \varphi\big((g^{-1})^m\big) \\ &= \big(\varphi(g^{-1})\big)^m \quad \text{(by the positive integer case applied to } g^{-1}\text{)} \\ &= \Big(\big(\varphi(g)\big)^{-1}\Big)^m \quad \text{(by part (ii))} \\ &= \big(\varphi(g)\big)^{-m} \\ &= \big(\varphi(g)\big)^n. \end{aligned}$$

(iv) Divisibility of Element Orders: Suppose $|g| = k < \infty$, which means $g^k = 1_G$ and k is the smallest positive integer with this property. Applying part (iii):

$$\big(\varphi(g)\big)^k = \varphi(g^k) = \varphi(1_G) = 1_H.$$

Let $m = |\varphi(g)|$ be the order of $\varphi(g)$ in H. By the division algorithm in $\mathbb{Z}$, write $k = qm + r$ with $0 \leq r < m$. Then:

$$1_H = \big(\varphi(g)\big)^k = \big(\varphi(g)\big)^{qm+r} = \Big(\big(\varphi(g)\big)^m\Big)^q \cdot \big(\varphi(g)\big)^r = (1_H)^q \cdot \big(\varphi(g)\big)^r = \big(\varphi(g)\big)^r.$$

If $r > 0$, this contradicts the minimality of $m = |\varphi(g)|$. Therefore, $r = 0$, which proves that $m \mid k$, i.e., $|\varphi(g)|$ divides $|g|$.

□

Definition 1.3 (Isomorphism and Automorphism)**.** A homomorphism $\varphi : G \to H$ is called an **isomorphism** if it is a bijection (both injective and surjective). If an isomorphism exists between G and H, we say G is **isomorphic** to H, denoted:

$$G \cong H. \tag{1.3}$$

An isomorphism from a group G to itself is called an **automorphism** of G. The set of all automorphisms of G forms a group under function composition, denoted $\mathrm{Aut}(G)$.

Definition 1.4 (Kernel and Image)**.** Let $\varphi : G \to H$ be a homomorphism.

(1) The **kernel** of φ is the pre-image of the identity element 1_H:

$$\ker(\varphi) = \{g \in G \mid \varphi(g) = 1_H\}. \tag{1.4}$$

(2) The **image** (or **range**) of φ is the set of all outputs:

$$\mathrm{im}(\varphi) = \varphi(G) = \{\varphi(g) \mid g \in G\} \subseteq H. \tag{1.5}$$

Proposition 1.5 (Injectivity and Kernel Characterization)**.** *Let $\varphi : G \to H$ be a group homomorphism.*

(i) $\ker(\varphi)$ *is a normal subgroup of G ($\ker(\varphi) \trianglelefteq G$).*

(ii) $\mathrm{im}(\varphi)$ *is a subgroup of H ($\mathrm{im}(\varphi) \leq H$).*

(iii) φ *is injective (one-to-one) if and only if* $\ker(\varphi) = \{1_G\}$.

Proof. **(i) Proof that** $\ker(\varphi) \trianglelefteq G$**:**

- *Non-empty:* By Proposition 1.2(i), $\varphi(1_G) = 1_H$, which implies $1_G \in \ker(\varphi)$. Thus $\ker(\varphi) \neq \emptyset$.
- *Closure under products and inverses:* Let $x, y \in \ker(\varphi)$. Then $\varphi(x) = 1_H$ and $\varphi(y) = 1_H$. Using the homomorphism property and Proposition 1.2(ii):

  $$\varphi(xy^{-1}) = \varphi(x)\varphi(y^{-1}) = \varphi(x)\big(\varphi(y)\big)^{-1} = 1_H \cdot (1_H)^{-1} = 1_H \cdot 1_H = 1_H.$$

  Hence $xy^{-1} \in \ker(\varphi)$, proving that $\ker(\varphi) \leq G$.
- *Normality (Conjugation Invariance):* Let $g \in G$ and $k \in \ker(\varphi)$. Then:

  $$\begin{aligned} \varphi(gkg^{-1}) &= \varphi(g)\varphi(k)\varphi(g^{-1}) \quad \text{(homomorphism property)} \\ &= \varphi(g) \cdot 1_H \cdot \big(\varphi(g)\big)^{-1} \quad (\text{since } k \in \ker\varphi) \\ &= \varphi(g)\big(\varphi(g)\big)^{-1} \\ &= 1_H. \end{aligned}$$

  Thus $gkg^{-1} \in \ker(\varphi)$ for all $g \in G$ and $k \in \ker(\varphi)$, which proves $g\ker(\varphi)g^{-1} \subseteq \ker(\varphi)$. Therefore, $\ker(\varphi) \trianglelefteq G$.

(ii) Proof that $\mathrm{im}(\varphi) \leq H$**:**

- *Non-empty:* Since $\varphi(1_G) = 1_H$, we have $1_H \in \mathrm{im}(\varphi)$, so $\mathrm{im}(\varphi) \neq \emptyset$.

- *Subgroup Criterion:* Let $h_1, h_2 \in \operatorname{im}(\varphi)$. By definition of the image, there exist elements $g_1, g_2 \in G$ such that $\varphi(g_1) = h_1$ and $\varphi(g_2) = h_2$. Then:

$$\begin{aligned} h_1 h_2^{-1} &= \varphi(g_1)(\varphi(g_2))^{-1} \\ &= \varphi(g_1)\varphi(g_2^{-1}) \quad \text{(by Proposition 1.2(ii))} \\ &= \varphi(g_1 g_2^{-1}) \quad \text{(homomorphism property).} \end{aligned}$$

Since $g_1, g_2 \in G$, $g_1 g_2^{-1} \in G$. Thus $h_1 h_2^{-1} = \varphi(g_1 g_2^{-1}) \in \operatorname{im}(\varphi)$. By the one-step subgroup criterion, $\operatorname{im}(\varphi) \leq H$.

(iii) Proof of Injectivity Criterion:

- **Direction ($\implies$):** Assume φ is injective. Let $x \in \ker(\varphi)$. Then $\varphi(x) = 1_H$. We already know from Proposition 1.2(i) that $\varphi(1_G) = 1_H$. Thus:

$$\varphi(x) = \varphi(1_G).$$

Since φ is injective, $\varphi(x) = \varphi(1_G) \implies x = 1_G$. Therefore, $\ker(\varphi) = \{1_G\}$.

- **Direction ($\impliedby$):** Assume $\ker(\varphi) = \{1_G\}$. Let $x, y \in G$ and suppose $\varphi(x) = \varphi(y)$. Multiplying both sides by $(\varphi(y))^{-1}$ in H:

$$\begin{aligned} \varphi(x)(\varphi(y))^{-1} = 1_H &\implies \varphi(x)\varphi(y^{-1}) = 1_H \\ &\implies \varphi(xy^{-1}) = 1_H \quad \text{(homomorphism property)} \\ &\implies xy^{-1} \in \ker(\varphi). \end{aligned}$$

Since $\ker(\varphi) = \{1_G\}$, this implies:

$$xy^{-1} = 1_G \implies x = y.$$

Hence φ is injective.

□

1.2 Subgroups and Subgroup Criteria

Definition 1.6 (Subgroup)**.** Let G be a group. A subset $H \subseteq G$ is called a **subgroup** of G, written $H \leq G$, if H is non-empty and forms a group in its own right under the binary operation inherited from G. If $H \leq G$ and $H \neq G$, we write $H < G$ (H is a proper subgroup).

Proposition 1.7 (One-Step Subgroup Criterion)**.** *A non-empty subset $H \subseteq G$ is a subgroup of G if and only if for all $x, y \in H$:*

$$xy^{-1} \in H. \tag{1.6}$$

Proof. • **Direction ($\implies$):** Suppose $H \leq G$. Let $x, y \in H$. Since H is a group, the inverse $y^{-1} \in H$. Since H is closed under multiplication, $x \in H$ and $y^{-1} \in H \implies xy^{-1} \in H$.

- **Direction ($\impliedby$):** Suppose $H \neq \emptyset$ and $xy^{-1} \in H$ for all $x, y \in H$. We verify the four group axioms for H:

1. *Identity:* Since $H \neq \emptyset$, there exists at least one element $x \in H$. Applying the hypothesis with $y = x$ gives:

$$1_G = xx^{-1} \in H.$$

Thus H contains the identity element of G.

2. *Inverses:* Let $y \in H$. Setting $x = 1_G \in H$ and applying the hypothesis gives:
$$y^{-1} = 1_G \cdot y^{-1} \in H.$$
Thus H is closed under taking inverses.
3. *Closure under multiplication:* Let $x, y \in H$. By step 2, $y^{-1} \in H$. Applying the hypothesis to x and y^{-1}:
$$xy = x(y^{-1})^{-1} \in H.$$
Thus H is closed under multiplication.
4. *Associativity:* The associative law $(ab)c = a(bc)$ holds for all elements in G, and therefore holds automatically for all elements in the subset H.

Therefore, H is a group under the operation of G, so $H \leq G$.

□

Proposition 1.8 (Two-Step Subgroup Criterion). *A non-empty subset $H \subseteq G$ is a subgroup if and only if:*

(1) *For all $x, y \in H$, $xy \in H$ (closure under multiplication), and*

(2) *For all $x \in H$, $x^{-1} \in H$ (closure under inverses).*

Proof.
- **Direction ($\implies$):** If $H \leq G$, then by definition of a group, H is closed under multiplication and inverses.
- **Direction ($\impliedby$):** Assume conditions (1) and (2) hold for $H \neq \emptyset$. Let $x, y \in H$. By condition (2), $y^{-1} \in H$. By condition (1) applied to x and y^{-1}, we have $xy^{-1} \in H$. By Proposition 1.7 (the One-Step Criterion), $H \leq G$.

□

Proposition 1.9 (Finite Subgroup Criterion). *Let H be a non-empty* ***finite*** *subset of a group G. Then $H \leq G$ if and only if H is closed under multiplication:*
$$\forall x, y \in H, \quad xy \in H. \tag{1.7}$$

Proof. If $H \leq G$, closure under multiplication is mandatory. Conversely, assume H is finite, non-empty, and closed under multiplication. By Proposition 1.8, it suffices to show that H is closed under taking inverses.

Let $x \in H$. Consider the infinite sequence of positive powers of x:
$$S = \{x, x^2, x^3, x^4, \dots\}.$$
Since H is closed under multiplication, $x \in H \implies x^2 = x \cdot x \in H \implies x^3 = x \cdot x^2 \in H$, and by induction, $x^k \in H$ for all $k \in \mathbb{Z}^+$. Thus $S \subseteq H$.

Since H is a finite set, the set S must also be finite. By the Pigeonhole Principle, the powers of x cannot all be distinct. Therefore, there exist positive integers $a > b \geq 1$ such that:
$$x^a = x^b.$$
Multiplying both sides of this equation by $(x^b)^{-1}$ in G:
$$x^a(x^b)^{-1} = x^b(x^b)^{-1} \implies x^{a-b} = 1_G.$$
Since $a > b$, we have $a - b \geq 1$, which proves that $1_G = x^{a-b} \in H$.

Now we isolate the inverse of x:
- If $a - b = 1$, then $x = 1_G$, so $x^{-1} = 1_G = x \in H$.
- If $a - b > 1$, then $a - b - 1 \geq 1$, so $x^{a-b-1} \in H$. Then:
$$x \cdot x^{a-b-1} = x^{a-b} = 1_G \implies x^{-1} = x^{a-b-1} \in H.$$

Thus $x^{-1} \in H$ for every $x \in H$. By Proposition 1.8, $H \leq G$. □

1.3 Centralizers, Normalizers, and the Center of a Group

Definition 1.10 (Centralizer and Normalizer)**.** Let G be a group and let $A \subseteq G$ be any subset.

(1) The **centralizer** of A in G, denoted $C_G(A)$, is the set of elements of G that commute with every element of A:

$$C_G(A) = \{g \in G \mid ga = ag \text{ for all } a \in A\}. \tag{1.8}$$

When $A = \{a\}$ is a singleton, we write $C_G(a)$.

(2) The **normalizer** of A in G, denoted $N_G(A)$, is the set of elements of G that stabilize A as a set under conjugation:

$$N_G(A) = \{g \in G \mid gAg^{-1} = A\}. \tag{1.9}$$

Definition 1.11 (Center of a Group)**.** The **center** of a group G, denoted $Z(G)$, is the centralizer of G in G:

$$Z(G) = C_G(G) = \{z \in G \mid zg = gz \text{ for all } g \in G\}. \tag{1.10}$$

Proposition 1.12 (Structural Properties of Centralizers, Normalizers, and Center)**.** *Let G be a group and $A \subseteq G$.*

(i) *$C_G(A) \leq G$.*

(ii) *$N_G(A) \leq G$.*

(iii) *$Z(G) \leq G$, $Z(G)$ is abelian, and $Z(G) \trianglelefteq G$.*

(iv) *$Z(G) \leq C_G(A) \leq N_G(A) \leq G$.*

(v) *G is abelian if and only if $Z(G) = G$.*

Proof. **(i) Proof that $C_G(A) \leq G$:**

- *Non-empty:* For every $a \in A$, $1_G \cdot a = a = a \cdot 1_G$. Thus $1_G \in C_G(A)$, so $C_G(A) \neq \emptyset$.
- *Subgroup Criterion:* Let $x, y \in C_G(A)$. For any $a \in A$, we have:

$$xa = ax \quad \text{and} \quad ya = ay.$$

Multiply $ya = ay$ on the left by y^{-1} and on the right by y^{-1}:

$$\begin{aligned} y^{-1}(ya)y^{-1} &= y^{-1}(ay)y^{-1} \\ (y^{-1}y)ay^{-1} &= y^{-1}a(yy^{-1}) \\ 1_G \cdot ay^{-1} &= y^{-1}a \cdot 1_G \\ ay^{-1} &= y^{-1}a. \end{aligned}$$

Now compute the action of xy^{-1} on a:

$$\begin{aligned} (xy^{-1})a &= x(y^{-1}a) && \text{(associativity)} \\ &= x(ay^{-1}) && \text{(since } y^{-1}a = ay^{-1}) \\ &= (xa)y^{-1} && \text{(associativity)} \\ &= (ax)y^{-1} && \text{(since } x \in C_G(A)) \\ &= a(xy^{-1}) && \text{(associativity).} \end{aligned}$$

Thus $xy^{-1} \in C_G(A)$. By Proposition 1.7, $C_G(A) \leq G$.

(ii) Proof that $N_G(A) \leq G$:

- *Non-empty:* $1_G A 1_G^{-1} = 1_G A 1_G = A$, so $1_G \in N_G(A)$.
- *Subgroup Criterion:* Let $x, y \in N_G(A)$. Then $xAx^{-1} = A$ and $yAy^{-1} = A$. From $yAy^{-1} = A$, multiply on the left by y^{-1} and on the right by y:

$$y^{-1}(yAy^{-1})y = y^{-1}Ay \implies (y^{-1}y)A(y^{-1}y) = y^{-1}Ay \implies A = y^{-1}Ay.$$

Now evaluate $(xy^{-1})A(xy^{-1})^{-1}$:

$$\begin{aligned}(xy^{-1})A(xy^{-1})^{-1} &= (xy^{-1})A((y^{-1})^{-1}x^{-1}) \\ &= (xy^{-1})A(yx^{-1}) \\ &= x(y^{-1}Ay)x^{-1} \\ &= xAx^{-1} \quad (\text{since } y^{-1}Ay = A) \\ &= A \quad (\text{since } xAx^{-1} = A).\end{aligned}$$

Thus $xy^{-1} \in N_G(A)$. By the one-step criterion, $N_G(A) \leq G$.

(iii) Proof that $Z(G) \trianglelefteq G$ is abelian: Setting $A = G$ in part (i) immediately gives $Z(G) = C_G(G) \leq G$. For any $z_1, z_2 \in Z(G)$, by definition z_1 commutes with every element of G, so $z_1 z_2 = z_2 z_1$. Thus $Z(G)$ is abelian.

To prove normality, let $g \in G$ and $z \in Z(G)$. Then:

$$\begin{aligned}gzg^{-1} &= (gz)g^{-1} \\ &= (zg)g^{-1} \quad (\text{since } z \in Z(G) \text{ commutes with } g) \\ &= z(gg^{-1}) \\ &= z \cdot 1_G = z.\end{aligned}$$

Since $z \in Z(G)$, $gzg^{-1} = z \in Z(G)$ for all $g \in G$. Thus $gZ(G)g^{-1} = Z(G)$, proving $Z(G) \trianglelefteq G$.

(iv) Proof of Inclusions:

- Let $z \in Z(G)$. Then $zg = gz$ for all $g \in G$. In particular, $za = az$ for all $a \in A \subseteq G$. Thus $z \in C_G(A)$, so $Z(G) \subseteq C_G(A)$.
- Let $g \in C_G(A)$. Then $ga = ag$ for all $a \in A$. Multiplying on the right by g^{-1} gives $gag^{-1} = a$ for all $a \in A$. Thus:

$$gAg^{-1} = \{gag^{-1} \mid a \in A\} = \{a \mid a \in A\} = A.$$

Hence $g \in N_G(A)$, so $C_G(A) \subseteq N_G(A)$.

Combining the subgroup results yields $Z(G) \leq C_G(A) \leq N_G(A) \leq G$.

(v) Abelian Criterion: G is abelian $\iff \forall x, y \in G, xy = yx \iff \forall x \in G, x \in C_G(G) = Z(G) \iff Z(G) = G$.

□

Theorem 1.13 (The N/C Theorem)**.** *Let $H \leq G$. Then $C_G(H) \trianglelefteq N_G(H)$, and the quotient group $N_G(H)/C_G(H)$ is isomorphic to a subgroup of* $\mathrm{Aut}(H)$*:*

$$N_G(H)/C_G(H) \hookrightarrow \mathrm{Aut}(H). \tag{1.11}$$

Proof. For each $g \in N_G(H)$, define the conjugation map:

$$\begin{aligned} \psi_g : H &\longrightarrow H \\ h &\longmapsto ghg^{-1}. \end{aligned}$$

Since $g \in N_G(H)$, $gHg^{-1} = H$, so ψ_g maps H bijectively onto H. Moreover:

$$\psi_g(h_1h_2) = g(h_1h_2)g^{-1} = (gh_1g^{-1})(gh_2g^{-1}) = \psi_g(h_1)\psi_g(h_2),$$

so $\psi_g \in \mathrm{Aut}(H)$. Now define the homomorphism:

$$\begin{aligned} \Psi : N_G(H) &\longrightarrow \mathrm{Aut}(H) \\ g &\longmapsto \psi_g. \end{aligned}$$

For any $g_1, g_2 \in N_G(H)$ and $h \in H$:

$$\psi_{g_1g_2}(h) = (g_1g_2)h(g_1g_2)^{-1} = g_1(g_2hg_2^{-1})g_1^{-1} = \psi_{g_1}(\psi_{g_2}(h)) = (\psi_{g_1} \circ \psi_{g_2})(h).$$

Thus $\Psi(g_1g_2) = \Psi(g_1) \circ \Psi(g_2)$, confirming Ψ is a group homomorphism. The kernel of Ψ is:

$$\begin{aligned} \ker(\Psi) &= \{g \in N_G(H) \mid \psi_g = \mathrm{id}_H\} \\ &= \{g \in N_G(H) \mid ghg^{-1} = h \text{ for all } h \in H\} \\ &= \{g \in N_G(H) \mid gh = hg \text{ for all } h \in H\} \\ &= N_G(H) \cap C_G(H) \\ &= C_G(H) \quad (\text{since } C_G(H) \leq N_G(H)). \end{aligned}$$

By Proposition 1.5, the kernel is a normal subgroup: $C_G(H) \trianglelefteq N_G(H)$. By the First Isomorphism Theorem (Theorem 2.5):

$$N_G(H)/C_G(H) \cong \mathrm{im}(\Psi) \leq \mathrm{Aut}(H).$$

□

1.4 Subgroups Generated by Subsets

Proposition 1.14 (Intersection of Subgroups)**.** *Let $\{H_i\}_{i\in I}$ be an arbitrary collection of subgroups of a group G. Then their intersection:*

$$H = \bigcap_{i\in I} H_i \tag{1.12}$$

is also a subgroup of G.

Proof. We verify the One-Step Subgroup Criterion:

1. *Non-empty:* Since each $H_i \leq G$, the identity element $1_G \in H_i$ for every $i \in I$. Therefore, $1_G \in \bigcap_{i\in I} H_i = H$, so $H \neq \emptyset$.

2. *Subgroup Criterion:* Let $x, y \in H$. By definition of intersection, $x \in H_i$ and $y \in H_i$ for every $i \in I$. Since each H_i is a subgroup of G, Proposition 1.7 guarantees:

$$xy^{-1} \in H_i \quad \text{for all } i \in I.$$

   Therefore, $xy^{-1} \in \bigcap_{i\in I} H_i = H$.

By Proposition 1.7, $H \leq G$. □

Definition 1.15 (Subgroup Generated by a Set)**.** Let A be a subset of a group G. The **subgroup generated by** A, denoted $\langle A \rangle$, is the intersection of all subgroups of G containing A:

$$\langle A \rangle = \bigcap_{\substack{H \leq G \\ A \subseteq H}} H. \tag{1.13}$$

If $A = \{a_1, \ldots, a_n\}$, we write $\langle a_1, \ldots, a_n \rangle$. If $G = \langle g \rangle$ for some $g \in G$, G is called a **cyclic group**.

Theorem 1.16 (Explicit Structure of Generated Subgroups)**.** *Let $A \subseteq G$ with $A \neq \emptyset$. Then $\langle A \rangle$ is the set of all finite products of elements of A and their inverses:*

$$\langle A \rangle = \bar{A} \coloneqq \{a_1^{\epsilon_1} a_2^{\epsilon_2} \cdots a_k^{\epsilon_k} \mid k \geq 0,\ a_i \in A,\ \epsilon_i \in \{+1, -1\}\}, \tag{1.14}$$

where $k = 0$ corresponds to the empty product 1_G.

Proof. We prove $\langle A \rangle = \bar{A}$ by double containment:

1. **Proof that $\bar{A} \leq G$:**
   - For $k = 0$, $1_G \in \bar{A}$, so $\bar{A} \neq \emptyset$.
   - Let $u = a_1^{\epsilon_1} \cdots a_k^{\epsilon_k} \in \bar{A}$ and $v = b_1^{\delta_1} \cdots b_m^{\delta_m} \in \bar{A}$. Then the inverse of v is:

   $$v^{-1} = (b_1^{\delta_1} \cdots b_m^{\delta_m})^{-1} = b_m^{-\delta_m} \cdots b_1^{-\delta_1} \in \bar{A},$$

   and the product is:

   $$uv^{-1} = a_1^{\epsilon_1} \cdots a_k^{\epsilon_k} b_m^{-\delta_m} \cdots b_1^{-\delta_1} \in \bar{A}.$$

   Thus $\bar{A} \leq G$ by the one-step criterion.

2. **Proof that $A \subseteq \bar{A}$:** For any $a \in A$, setting $k = 1, a_1 = a, \epsilon_1 = 1$ gives $a = a^1 \in \bar{A}$. Thus $A \subseteq \bar{A}$.

3. **Proof that $\langle A \rangle \subseteq \bar{A}$:** Since $\bar{A}$ is a subgroup of G containing A, $\bar{A}$ is one of the subgroups in the intersection defining $\langle A \rangle$. Since an intersection is contained in each of its terms:

   $$\langle A \rangle = \bigcap_{\substack{H \leq G \\ A \subseteq H}} H \subseteq \bar{A}.$$

4. **Proof that $\bar{A} \subseteq \langle A \rangle$:** Let H be any subgroup of G containing A. Since H is closed under taking inverses, $a \in A \implies a^{-1} \in H$. Since H is closed under multiplication, any finite product $a_1^{\epsilon_1} \cdots a_k^{\epsilon_k}$ with $a_i \in A, \epsilon_i = \pm 1$ must lie in H. Thus $\bar{A} \subseteq H$ for every subgroup H containing A. Taking the intersection over all such H:

   $$\bar{A} \subseteq \bigcap_{\substack{H \leq G \\ A \subseteq H}} H = \langle A \rangle.$$

Combining $\langle A \rangle \subseteq \bar{A}$ and $\bar{A} \subseteq \langle A \rangle$ gives $\langle A \rangle = \bar{A}$. □

1.5 Cosets, Lagrange's Theorem, and Number-Theoretic Applications

Definition 1.17 (Left and Right Cosets)**.** Let $H \leq G$ and $g \in G$.

(1) The **left coset** of H in G containing g is:

$$gH = \{gh \mid h \in H\}. \tag{1.15}$$

(2) The **right coset** of H in G containing g is:

$$Hg = \{hg \mid h \in H\}. \tag{1.16}$$

The element g is called a **coset representative** of gH (or Hg).

Lemma 1.18 (Coset Partition and Equivalence Relation Properties)**.** *Let $H \leq G$.*

(i) *The relation $\sim_L$ defined by $x \sim_L y \iff x^{-1}y \in H$ is an equivalence relation on G.*

(ii) *The equivalence class of $x \in G$ under $\sim_L$ is the left coset xH.*

(iii) *For any $x, y \in G$, the following statements are logically equivalent:*

- ***(a)*** $xH = yH$,
- ***(b)*** $y^{-1}x \in H$,
- ***(c)*** $x \in yH$,
- ***(d)*** $xH \cap yH \neq \emptyset$.

(iv) *The left cosets of H in G form a partition of G.*

(v) *For every $g \in G$, $|gH| = |H| = |Hg|$.*

Proof. **(i) Proof of Equivalence Relation:**

- *Reflexive:* For all $x \in G$, $x^{-1}x = 1_G \in H$ (since $H \leq G$). Thus $x \sim_L x$.
- *Symmetric:* Suppose $x \sim_L y$, so $x^{-1}y \in H$. Since H is closed under inverses:

$$(x^{-1}y)^{-1} = y^{-1}(x^{-1})^{-1} = y^{-1}x \in H.$$

  Thus $y \sim_L x$.
- *Transitive:* Suppose $x \sim_L y$ and $y \sim_L z$, so $x^{-1}y \in H$ and $y^{-1}z \in H$. Since H is closed under multiplication:

$$(x^{-1}y)(y^{-1}z) = x^{-1}(yy^{-1})z = x^{-1}(1_G)z = x^{-1}z \in H.$$

  Thus $x \sim_L z$.

(ii) Equivalence Class Structure: The equivalence class of x is:

$$\begin{aligned} [x]_{\sim_L} &= \{y \in G \mid x \sim_L y\} = \{y \in G \mid x^{-1}y \in H\} \\ &= \{y \in G \mid x^{-1}y = h \text{ for some } h \in H\} \\ &= \{y \in G \mid y = xh \text{ for some } h \in H\} \\ &= xH. \end{aligned}$$

(iii) Equivalence of Coset Conditions:

- $(a) \implies (c)$**:** If $xH = yH$, then since $1_G \in H$, $x = x \cdot 1_G \in xH = yH$.
- $(c) \implies (b)$**:** If $x \in yH$, then $x = yh$ for some $h \in H$. Multiplying on the left by y^{-1} gives $y^{-1}x = h \in H$.
- $(b) \implies (a)$**:** If $y^{-1}x = h \in H$, then $x = yh$. For any $h' \in H$, $xh' = (yh)h' = y(hh') \in yH$, so $xH \subseteq yH$. Moreover, $x = yh \implies y = xh^{-1}$. Since $h \in H \implies h^{-1} \in H$, for any $h'' \in H$, $yh'' = x(h^{-1}h'') \in xH$, so $yH \subseteq xH$. Thus $xH = yH$.
- $(a) \implies (d)$**:** If $xH = yH$, then $xH \cap yH = xH \neq \emptyset$ since $x \in xH$.
- $(d) \implies (a)$**:** Suppose $z \in xH \cap yH$. Then $z = xh_1 = yh_2$ for some $h_1, h_2 \in H$. Then $x = yh_2h_1^{-1}$. Since $h_2h_1^{-1} \in H$, $x \in yH$, which by $(c) \implies (a)$ gives $xH = yH$.

(iv) Partition of G: Since $\sim_L$ is an equivalence relation, its equivalence classes (which are precisely the left cosets gH) form a partition of G.

(v) Equal Cardinality: For any $g \in G$, define the map $\lambda_g : H \to gH$ by $\lambda_g(h) = gh$.

- *Surjective:* Every element of gH is of the form $gh = \lambda_g(h)$ for some $h \in H$.
- *Injective:* Suppose $\lambda_g(h_1) = \lambda_g(h_2)$. Then $gh_1 = gh_2$. Multiplying on the left by g^{-1}:

$$g^{-1}(gh_1) = g^{-1}(gh_2) \implies (g^{-1}g)h_1 = (g^{-1}g)h_2 \implies h_1 = h_2.$$

Thus λ_g is a bijection, which proves $|gH| = |H|$. A completely symmetric argument with $\rho_g : H \to Hg$ defined by $\rho_g(h) = hg$ shows $|Hg| = |H|$.

□

Definition 1.19 (Index of a Subgroup)**.** Let $H \leq G$. The **index** of H in G, denoted $[G : H]$, is the number of distinct left (or right) cosets of H in G.

Theorem 1.20 (Lagrange's Theorem)**.** *If G is a finite group and H is a subgroup of G, then the order of H divides the order of G, and:*

$$|G| = [G : H] \cdot |H|. \tag{1.17}$$

Proof. Let $k = [G : H]$ be the number of distinct left cosets of H in G, and choose representatives $g_1, g_2, \ldots, g_k \in G$ such that:

$$g_1H,\ g_2H,\ \ldots,\ g_kH$$

are the distinct pairwise disjoint left cosets of H. By Lemma 1.18(iv), these cosets partition G:

$$G = g_1H \cup g_2H \cup \cdots \cup g_kH \quad \text{with } g_iH \cap g_jH = \emptyset \text{ for } i \neq j.$$

Taking the cardinality of both sides:

$$|G| = |g_1H| + |g_2H| + \cdots + |g_kH|.$$

By Lemma 1.18(v), each coset satisfies $|g_iH| = |H|$:

$$|G| = \sum_{i=1}^{k} |H| = k \cdot |H| = [G : H] \cdot |H|.$$

Since $[G : H]$ is an integer, $|H|$ divides $|G|$. □

Corollary 1.21 (Order of an Element Divides Group Order)**.** *Let G be a finite group and let $g \in G$. Then the order of g divides $|G|$, and:*

$$g^{|G|} = 1_G. \tag{1.18}$$

Proof. Consider the cyclic subgroup generated by g: $H = \langle g \rangle = \{1_G, g, g^2, \dots, g^{|g|-1}\}$. The order of H is $|H| = |g|$. By Lagrange's Theorem (Theorem 1.20), $|H|$ divides $|G|$, so $|g|$ divides $|G|$.

Let $|G| = m \cdot |g|$ for some positive integer $m \geq 1$. Then:

$$g^{|G|} = g^{|g| \cdot m} = \left(g^{|g|}\right)^m = (1_G)^m = 1_G.$$

□

Corollary 1.22 (Groups of Prime Order are Cyclic)**.** *If G is a group of prime order p, then G is cyclic and isomorphic to $\mathbb{Z}/p\mathbb{Z}$. Moreover, every non-identity element of G is a generator.*

Proof. Let $x \in G$ with $x \neq 1_G$. Consider the cyclic subgroup $\langle x \rangle \leq G$. Since $x \neq 1_G$, $|\langle x \rangle| \geq 2$. By Lagrange's Theorem, $|\langle x \rangle|$ must divide $|G| = p$. Since p is prime, its only positive divisors are 1 and p. Since $|\langle x \rangle| > 1$, we must have $|\langle x \rangle| = p$. Therefore, $\langle x \rangle = G$, which proves G is cyclic. Any cyclic group of order p is isomorphic to $\mathbb{Z}/p\mathbb{Z}$ via the isomorphism $x^k \mapsto \bar{k}$. □

1.5.1 Applications to Number Theory

Theorem 1.23 (Euler's Totient Theorem)**.** *Let $n \in \mathbb{Z}^+$ and let $a \in \mathbb{Z}$ with $\gcd(a, n) = 1$. Then:*

$$a^{\phi(n)} \equiv 1 \pmod{n}, \tag{1.19}$$

where $\phi(n) = |(\mathbb{Z}/n\mathbb{Z})^\times|$ is Euler's totient function.

Proof. Consider the set of units in the ring $\mathbb{Z}/n\mathbb{Z}$:

$$G = (\mathbb{Z}/n\mathbb{Z})^\times = \{\bar{k} \in \mathbb{Z}/n\mathbb{Z} \mid \gcd(k, n) = 1\}.$$

Under multiplication modulo n, G forms an abelian group of order $|G| = \phi(n)$. Since $\gcd(a, n) = 1$, the residue class $\bar{a} = a \pmod{n}$ is an element of G. Applying Corollary 1.21 to the finite group G:

$$\bar{a}^{|G|} = \bar{a}^{\phi(n)} = \bar{1} \in (\mathbb{Z}/n\mathbb{Z})^\times.$$

In modular arithmetic notation, this translates directly to:

$$a^{\phi(n)} \equiv 1 \pmod{n}.$$

□

Theorem 1.24 (Fermat's Little Theorem)**.** *Let p be a prime number. For any $a \in \mathbb{Z}$:*

$$a^p \equiv a \pmod{p}. \tag{1.20}$$

In particular, if $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

Proof. For a prime p, the integers coprime to p in $\{1, 2, \dots, p-1\}$ are all $p - 1$ elements, so $\phi(p) = p - 1$. If $p \nmid a$, then $\gcd(a, p) = 1$. Applying Euler's Totient Theorem with $n = p$:

$$a^{\phi(p)} = a^{p-1} \equiv 1 \pmod{p}.$$

Multiplying both sides by a gives:

$$a^p \equiv a \pmod{p}.$$

If $p \mid a$, then $a \equiv 0 \pmod{p}$, which gives $a^p \equiv 0^p \equiv 0 \equiv a \pmod{p}$, so the identity holds unconditionally for all $a \in \mathbb{Z}$. □

Theorem 1.25 (Wilson's Theorem)**.** *An integer $p \geq 2$ is prime if and only if:*

$$(p-1)! \equiv -1 \pmod{p}. \tag{1.21}$$

Proof.

- **Direction ($\Longrightarrow$):** Assume p is prime. For $p = 2$, $(2-1)! = 1! = 1 \equiv -1 \pmod{2}$. Now assume $p > 2$ (so p is odd). Consider the group of units $G = (\mathbb{Z}/p\mathbb{Z})^\times = \{\bar{1}, \bar{2}, \dots, \overline{p-1}\}$. Every element $\bar{x} \in G$ has a unique multiplicative inverse $\bar{x}^{-1} \in G$.

  An element is its own multiplicative inverse if and only if:

$$\begin{aligned} x^2 \equiv 1 \pmod{p} &\iff x^2 - 1 \equiv 0 \pmod{p} \\ &\iff (x-1)(x+1) \equiv 0 \pmod{p} \\ &\iff p \mid (x-1)(x+1). \end{aligned}$$

  By Euclid's Lemma, since p is prime:

$$p \mid (x-1) \implies x \equiv 1 \pmod{p}, \quad \text{or} \quad p \mid (x+1) \implies x \equiv -1 \equiv p-1 \pmod{p}.$$

  Thus, among the $p-1$ elements of G, exactly two elements (1 and $p-1$) are self-inverse. The remaining $p-3$ elements:

$$\{2, 3, \dots, p-2\}$$

  can be paired up into $m = \frac{p-3}{2}$ disjoint pairs of mutual inverses (x_j, y_j) with $x_j y_j \equiv 1 \pmod{p}$.

  Taking the product of all elements in G:

$$\begin{aligned} (p-1)! &= 1 \cdot \left(\prod_{j=1}^{(p-3)/2} x_j y_j \right) \cdot (p-1) \\ &\equiv 1 \cdot (1 \cdot 1 \cdots 1) \cdot (p-1) \pmod{p} \\ &\equiv p-1 \pmod{p} \\ &\equiv -1 \pmod{p}. \end{aligned}$$

- **Direction ($\Longleftarrow$):** Assume $(p-1)! \equiv -1 \pmod{p}$. Suppose for contradiction that p is composite, with a proper divisor d such that $1 < d < p$. Since $d \leq p-1$, d appears as one of the factors in $(p-1)! = 1 \cdot 2 \cdots d \cdots (p-1)$. Therefore:

$$d \mid (p-1)!.$$

  By assumption, $(p-1)! + 1 = kp$ for some integer k. Since $d \mid p$, $d \mid kp$, so $d \mid ((p-1)!+1)$. Since d divides both $(p-1)!$ and $(p-1)!+1$, d must divide their difference:

$$d \mid \Big(((p-1)! + 1) - (p-1)! \Big) = 1.$$

  This forces $d = 1$, contradicting $1 < d < p$. Therefore, p must be prime.

□

1.6 Exercises and Step-by-Step Solutions

Problem 1.26 (Exercise 1: Inverse of an Isomorphism)**.** Prove that if $\varphi : G \to H$ is an isomorphism of groups, then its set-theoretic inverse mapping $\varphi^{-1} : H \to G$ is also an isomorphism of groups.

Solution. Since $\varphi : G \to H$ is an isomorphism, it is a bijective homomorphism. Being a bijection, the inverse mapping $\varphi^{-1} : H \to G$ exists and is also a bijection. It remains only to prove that φ^{-1} is a group homomorphism.

Let $y_1, y_2 \in H$. Since φ is surjective, there exist unique elements $x_1, x_2 \in G$ such that:

$$\varphi(x_1) = y_1 \iff x_1 = \varphi^{-1}(y_1) \quad \text{and} \quad \varphi(x_2) = y_2 \iff x_2 = \varphi^{-1}(y_2).$$

Using the fact that φ is a homomorphism:

$$\varphi(x_1x_2) = \varphi(x_1)\varphi(x_2) = y_1y_2.$$

Applying φ^{-1} to both sides:

$$\begin{aligned}\varphi^{-1}(y_1y_2) &= \varphi^{-1}(\varphi(x_1x_2)) \\ &= x_1x_2 \quad (\text{since } \varphi^{-1} \circ \varphi = \mathrm{id}_G) \\ &= \varphi^{-1}(y_1)\varphi^{-1}(y_2).\end{aligned}$$

Thus φ^{-1} preserves the group operation. Being a bijective homomorphism, φ^{-1} is an isomorphism. □

Problem 1.27 (Exercise 2: The Inversion Map)**.** Let G be a group. Prove that the inversion map $f : G \to G$ defined by $f(g) = g^{-1}$ is an automorphism of G if and only if G is abelian.

Solution. First, note that f is always a bijection on G:

- $f(f(g)) = (g^{-1})^{-1} = g$, so $f \circ f = \mathrm{id}_G$, meaning f is its own inverse and hence bijective.

Thus f is an automorphism if and only if f is a homomorphism.

We test the homomorphism condition for all $x, y \in G$:

$$f(xy) = (xy)^{-1} = y^{-1}x^{-1}.$$

On the other hand:

$$f(x)f(y) = x^{-1}y^{-1}.$$

Therefore:

$$\begin{aligned} f \text{ is a homomorphism} &\iff \forall x, y \in G,\ f(xy) = f(x)f(y) \\ &\iff \forall x, y \in G,\ y^{-1}x^{-1} = x^{-1}y^{-1} \\ &\iff \forall x, y \in G,\ (xy)^{-1} = (yx)^{-1} \\ &\iff \forall x, y \in G,\ xy = yx \quad (\text{taking inverses of both sides}) \\ &\iff G \text{ is abelian.}\end{aligned}$$

□

Problem 1.28 (Exercise 3: Union of Subgroups)**.** Let H and K be subgroups of a group G. Prove that the union $H \cup K$ is a subgroup of G if and only if $H \subseteq K$ or $K \subseteq H$.

Solution.

- **Direction ($\Longleftarrow$):** If $H \subseteq K$, then $H \cup K = K$, which is a subgroup since $K \leq G$. If $K \subseteq H$, then $H \cup K = H \leq G$.

- **Direction ($\Longrightarrow$):** Suppose $H \cup K \leq G$. Assume for contradiction that $H \not\subseteq K$ and $K \not\subseteq H$. Then there exist elements:

$$h \in H \setminus K \quad \text{and} \quad k \in K \setminus H.$$

Since $h \in H \subseteq H \cup K$ and $k \in K \subseteq H \cup K$, and since $H \cup K$ is a subgroup, the product:

$$hk \in H \cup K.$$

This means that either $hk \in H$ or $hk \in K$:

– *Case 1:* If $hk \in H$, then since $h \in H$, $h^{-1} \in H$. Thus $k = h^{-1}(hk) \in H$, which contradicts $k \notin H$.
– *Case 2:* If $hk \in K$, then since $k \in K$, $k^{-1} \in K$. Thus $h = (hk)k^{-1} \in K$, which contradicts $h \notin K$.

In both cases we reach a contradiction. Therefore, we must have $H \subseteq K$ or $K \subseteq H$.

□

Problem 1.29 (Exercise 4: Subgroups of Index 2)**.** Show that if G is a group and $H \leq G$ with index $[G : H] = 2$, then $gH = Hg$ for every $g \in G$ (so H is a normal subgroup).

Solution. Since $[G : H] = 2$, there are exactly two distinct left cosets and two distinct right cosets of H in G. One coset is always $1_G H = H = H1_G$.

Let $g \in G$.

- *Case 1:* $g \in H$. Then $gH = H = Hg$.
- *Case 2:* $g \notin H$. Since left cosets partition G, the two left cosets are H and gH. Thus:

$$G = H \cup gH \quad \text{(disjoint union)} \implies gH = G \setminus H.$$

Similarly, the two right cosets are H and Hg, so:

$$G = H \cup Hg \quad \text{(disjoint union)} \implies Hg = G \setminus H.$$

Therefore, $gH = G \setminus H = Hg$.

In all cases, $gH = Hg$, which proves $H \trianglelefteq G$. □

Problem 1.30 (Exercise 5: Center as Intersection of Centralizers)**.** Prove that for any group G, the center is the intersection of all element centralizers:

$$Z(G) = \bigcap_{x \in G} C_G(x). \tag{1.22}$$

Solution. We prove double containment:

- **Direction** ($\subseteq$)**:** Let $z \in Z(G)$. By definition of the center, $zg = gz$ for all $g \in G$. In particular, for any fixed $x \in G$, $zx = xz$, which means $z \in C_G(x)$. Since this holds for every $x \in G$:

$$z \in \bigcap_{x \in G} C_G(x) \implies Z(G) \subseteq \bigcap_{x \in G} C_G(x).$$

- **Direction** ($\supseteq$)**:** Let $g \in \bigcap_{x \in G} C_G(x)$. By definition of intersection, $g \in C_G(x)$ for every $x \in G$. By definition of the centralizer $C_G(x)$, this means $gx = xg$ for every $x \in G$. Thus g commutes with all elements of G, which by definition of the center means $g \in Z(G)$. Therefore:

$$\bigcap_{x \in G} C_G(x) \subseteq Z(G).$$

Thus $Z(G) = \bigcap_{x \in G} C_G(x)$. □

Problem 1.31 (Exercise 6: Wilson's Theorem and Quadratic Residues)**.** Use Wilson's Theorem to prove that if p is a prime of the form $p = 4k + 1$, then -1 is a quadratic residue modulo p, and specifically:

$$\left[\left(\frac{p-1}{2}\right)!\right]^2 \equiv -1 \pmod{p}. \tag{1.23}$$

Solution. Let $m = \frac{p-1}{2} = 2k$. By Wilson's Theorem (Theorem 1.25):

$$(p-1)! = 1 \cdot 2 \cdots m \cdot (m+1) \cdots (p-1) \equiv -1 \pmod{p}.$$

We rewrite the factors from $m+1$ to $p-1$ modulo p:

$$\begin{aligned} p-1 &\equiv -1 \pmod{p}, \\ p-2 &\equiv -2 \pmod{p}, \\ &\vdots \\ p-j &\equiv -j \pmod{p}, \\ &\vdots \\ m+1 = p-m &\equiv -m \pmod{p}. \end{aligned}$$

There are m such factors. Substituting these into the factorial product:

$$\begin{aligned} (p-1)! &= \left(\prod_{j=1}^{m} j\right) \cdot \left(\prod_{j=1}^{m} (p-j)\right) \\ &\equiv (m!) \cdot \left(\prod_{j=1}^{m} (-j)\right) \pmod{p} \\ &\equiv (m!) \cdot (-1)^m (m!) \pmod{p} \\ &\equiv (-1)^m (m!)^2 \pmod{p}. \end{aligned}$$

Since $p = 4k+1$, $m = \frac{p-1}{2} = 2k$ is an **even** integer. Therefore:

$$(-1)^m = (-1)^{2k} = 1.$$

Substituting this into the congruence:

$$(p-1)! \equiv 1 \cdot (m!)^2 \equiv (m!)^2 \pmod{p}.$$

Since $(p-1)! \equiv -1 \pmod{p}$ by Wilson's Theorem:

$$\left[\left(\frac{p-1}{2}\right)!\right]^2 \equiv -1 \pmod{p}.$$

Setting $x = \left(\frac{p-1}{2}\right)!$, we have $x^2 \equiv -1 \pmod{p}$, proving that -1 is a quadratic residue modulo p. □

Chapter 2

Quotient Groups, Isomorphism Theorems, and Direct Products

In this chapter, we develop the core machinery of modern group theory: normal subgroups, quotient (factor) groups, the four classical Isomorphism Theorems, the direct product of groups, and the commutator subgroup with its associated abelianization. Every theorem is proved in complete, rigorous detail.

2.1 Normal Subgroups and Quotient Groups

Definition 2.1 (Normal Subgroup)**.** A subgroup N of a group G is called a **normal subgroup**, denoted $N \trianglelefteq G$, if for every $g \in G$ and every $n \in N$:

$$gng^{-1} \in N. \tag{2.1}$$

Equivalently, $gNg^{-1} = N$ for all $g \in G$, or $gN = Ng$ for all $g \in G$.

Proposition 2.2 (Comprehensive Normality Criteria)**.** *Let N be a subgroup of G ($N \leq G$). The following statements are logically equivalent:*

(1) *$N \trianglelefteq G$ (i.e., $gng^{-1} \in N$ for all $g \in G$ and $n \in N$).*

(2) *$gNg^{-1} = N$ for all $g \in G$.*

(3) *$N_G(N) = G$.*

(4) *$gN = Ng$ for all $g \in G$ (every left coset equals the corresponding right coset).*

(5) *The set of left cosets $G/N = \{gN \mid g \in G\}$ forms a group under the operation:*

$$(aN)(bN) = (ab)N. \tag{2.2}$$

(6) *N is the kernel of some group homomorphism $\varphi : G \to H$.*

Proof. We establish the circular chain of implications (1) $\implies$ (2) $\implies$ (3) $\implies$ (4) $\implies$ (5) $\implies$ (6) $\implies$ (1):

- **Implication** (1) $\implies$ (2)**:** Assume $gng^{-1} \in N$ for all $g \in G$ and $n \in N$. This means $gNg^{-1} \subseteq N$ for all $g \in G$. Replacing g with g^{-1} (which also belongs to G):

  $$g^{-1}N(g^{-1})^{-1} \subseteq N \implies g^{-1}Ng \subseteq N.$$

  Multiplying on the left by g and on the right by g^{-1}:

  $$g(g^{-1}Ng)g^{-1} \subseteq gNg^{-1} \implies (gg^{-1})N(gg^{-1}) \subseteq gNg^{-1} \implies N \subseteq gNg^{-1}.$$

  Since $gNg^{-1} \subseteq N$ and $N \subseteq gNg^{-1}$, we have $gNg^{-1} = N$ for all $g \in G$.

- **Implication** (2) $\implies$ (3)**:** The normalizer of N in G is defined as $N_G(N) = \{g \in G \mid gNg^{-1} = N\}$. If $gNg^{-1} = N$ for all $g \in G$, then every element of G belongs to $N_G(N)$. Thus $N_G(N) = G$.

- **Implication** (3) $\implies$ (4)**:** If $N_G(N) = G$, then for every $g \in G$, $gNg^{-1} = N$. Multiplying both sides on the right by g:

$$(gNg^{-1})g = Ng \implies gN(g^{-1}g) = Ng \implies gN = Ng.$$

- **Implication** (4) $\implies$ (5)**:** Suppose $gN = Ng$ for all $g \in G$. We must prove that coset multiplication $(aN)(bN) = (ab)N$ is well-defined and satisfies the group axioms.

  Well-definedness: Let $aN = a'N$ and $bN = b'N$. Then $a' = an_1$ and $b' = bn_2$ for some $n_1, n_2 \in N$. We must show that $(a'b')N = (ab)N$. Compute $a'b'$:

$$a'b' = (an_1)(bn_2) = a(n_1b)n_2.$$

  Since $n_1 \in N$ and $Nb = bN$, we have $n_1b \in Nb = bN$, so there exists $n_3 \in N$ such that $n_1b = bn_3$. Substituting this in:

$$a'b' = a(bn_3)n_2 = (ab)(n_3n_2).$$

  Since $n_3, n_2 \in N$ and N is a subgroup, $n_3n_2 \in N$. Therefore, $a'b' \in (ab)N$, which implies $(a'b')N = (ab)N$. Thus the operation is well-defined.

  Group Axioms for G/N:

  a. *Associativity:* For all $aN, bN, cN \in G/N$:

$$\begin{aligned}((aN)(bN))(cN) &= ((ab)N)(cN) = ((ab)c)N \\ &= (a(bc))N = (aN)((bc)N) = (aN)((bN)(cN)).\end{aligned}$$

  b. *Identity:* $(1_GN)(aN) = (1_Ga)N = aN = (a \cdot 1_G)N = (aN)(1_GN)$. Thus $1_GN = N$ is the identity element.

  c. *Inverses:* For any $aN \in G/N$, $(a^{-1}N)(aN) = (a^{-1}a)N = N = (aa^{-1})N = (aN)(a^{-1}N)$. Thus $(aN)^{-1} = a^{-1}N$.

  Thus G/N is a group.

- **Implication** (5) $\implies$ (6)**:** Assume G/N is a group. Define the canonical projection map:

$$\begin{aligned}\pi : G &\longrightarrow G/N \\ g &\longmapsto gN.\end{aligned}$$

  For any $x, y \in G$, $\pi(xy) = (xy)N = (xN)(yN) = \pi(x)\pi(y)$. Thus π is a group homomorphism. The kernel of π is:

$$\begin{aligned}\ker(\pi) &= \{g \in G \mid \pi(g) = 1_{G/N}\} \\ &= \{g \in G \mid gN = N\} \\ &= \{g \in G \mid g \in N\} \\ &= N.\end{aligned}$$

  Thus $N = \ker(\pi)$, so N is the kernel of a homomorphism.

- **Implication** (6) $\implies$ (1)**:** Suppose $N = \ker(\varphi)$ for some homomorphism $\varphi : G \to H$. By Proposition 1.5(i), $\ker(\varphi) \trianglelefteq G$. Thus $N \trianglelefteq G$.

□

Definition 2.3 (Quotient Group)**.** Let $N \trianglelefteq G$. The group $G/N = \{gN \mid g \in G\}$ with operation $(aN)(bN) = (ab)N$ is called the **quotient group** (or **factor group**) of G by N. By Lagrange's Theorem, if G is finite:

$$|G/N| = [G : N] = \frac{|G|}{|N|}. \tag{2.3}$$

Proposition 2.4 (Subgroups of Index 2 are Normal)**.** *Let G be a group. If $H \leq G$ and $[G : H] = 2$, then $H \trianglelefteq G$.*

Proof. Since $[G : H] = 2$, the set of left cosets consists of exactly two cosets: H and gH (where $g \notin H$). Since left cosets partition G:

$$G = H \cup gH \quad \text{(disjoint union)} \implies gH = G \setminus H.$$

Similarly, the right cosets of H in G are H and Hg for $g \notin H$. Since right cosets also partition G:

$$G = H \cup Hg \quad \text{(disjoint union)} \implies Hg = G \setminus H.$$

Therefore, $gH = G \setminus H = Hg$ for all $g \notin H$. For $g \in H$, $gH = H = Hg$. Thus $gH = Hg$ for all $g \in G$. By Proposition 2.2(4), $H \trianglelefteq G$. □

2.2 The Four Isomorphism Theorems

Theorem 2.5 (First Isomorphism Theorem / Fundamental Homomorphism Theorem)**.** *Let $\varphi : G \to H$ be a group homomorphism. Then:*

$$\ker(\varphi) \trianglelefteq G \quad \text{and} \quad G/\ker(\varphi) \cong \operatorname{im}(\varphi). \tag{2.4}$$

Specifically, the map:

$$\bar{\varphi} : G/\ker(\varphi) \longrightarrow \operatorname{im}(\varphi) \tag{2.5}$$
$$g\ker(\varphi) \longmapsto \varphi(g) \tag{2.6}$$

is a well-defined group isomorphism.

Proof. Let $K = \ker(\varphi)$. By Proposition 1.5(i), $K \trianglelefteq G$, so the quotient group G/K is well-defined. We verify the four essential properties of $\bar{\varphi}$:

1. **Well-Definedness:** Suppose $aK = bK$ in G/K. We must show $\bar{\varphi}(aK) = \bar{\varphi}(bK)$, i.e., $\varphi(a) = \varphi(b)$. By Lemma 1.18(iii), $aK = bK \implies b^{-1}a \in K = \ker(\varphi)$. Applying φ:

$$\begin{aligned} \varphi(b^{-1}a) = 1_H &\implies \varphi(b^{-1})\varphi(a) = 1_H \\ &\implies \big(\varphi(b)\big)^{-1}\varphi(a) = 1_H \\ &\implies \varphi(a) = \varphi(b) \\ &\implies \bar{\varphi}(aK) = \bar{\varphi}(bK). \end{aligned}$$

   Thus $\bar{\varphi}$ is well-defined (independent of the coset representative).

2. **Homomorphism Property:** For any two cosets $aK, bK \in G/K$:

$$\begin{aligned} \bar{\varphi}((aK)(bK)) &= \bar{\varphi}((ab)K) \quad \text{(definition of quotient group multiplication)} \\ &= \varphi(ab) \quad \text{(definition of } \bar{\varphi}) \\ &= \varphi(a)\varphi(b) \quad \text{(since } \varphi \text{ is a homomorphism)} \\ &= \bar{\varphi}(aK)\bar{\varphi}(bK) \quad \text{(definition of } \bar{\varphi}). \end{aligned}$$

   Thus $\bar{\varphi}$ is a group homomorphism.

3. **Injectivity (Kernel Triviality):** We compute the kernel of $\bar{\varphi}$:

$$\begin{aligned}\ker(\bar{\varphi}) &= \{gK \in G/K \mid \bar{\varphi}(gK) = 1_H\} \\ &= \{gK \in G/K \mid \varphi(g) = 1_H\} \\ &= \{gK \in G/K \mid g \in \ker(\varphi)\} \\ &= \{gK \in G/K \mid g \in K\} \\ &= \{1_G K\} = \{K\}.\end{aligned}$$

   Since the kernel of $\bar{\varphi}$ contains only the identity element $K = 1_{G/K}$ of the quotient group G/K, Proposition 1.5(iii) implies that $\bar{\varphi}$ is injective.

4. **Surjectivity:** Let $h \in \operatorname{im}(\varphi)$. By definition of the image, there exists some element $g \in G$ such that $\varphi(g) = h$. Consider the coset $gK \in G/K$. Then:

$$\bar{\varphi}(gK) = \varphi(g) = h.$$

   Thus every element of $\operatorname{im}(\varphi)$ is mapped to by $\bar{\varphi}$, so $\bar{\varphi}$ is surjective.

Since $\bar{\varphi}$ is a well-defined bijective homomorphism, it is a group isomorphism:

$$G/\ker(\varphi) \cong \operatorname{im}(\varphi).$$

□

Theorem 2.6 (Second Isomorphism Theorem / Diamond Isomorphism Theorem)**.** *Let G be a group, let $A \leq G$, and let $B \trianglelefteq G$. Then:*

(i) $AB = \{ab \mid a \in A, b \in B\} \leq G$.

(ii) $B \trianglelefteq AB$.

(iii) $A \cap B \trianglelefteq A$.

(iv) $AB/B \cong A/(A \cap B)$.

Proof. **(i) Proof that $AB \leq G$:** We show that $AB = BA$. Let $a \in A$ and $b \in B$. Since $B \trianglelefteq G$, $a^{-1}ba \in B$. Then:

$$ba = a(a^{-1}ba) \in AB \implies BA \subseteq AB.$$

Similarly, $aba^{-1} \in B$, so $ab = (aba^{-1})a \in BA \implies AB \subseteq BA$. Thus $AB = BA$.

Now use the One-Step Subgroup Criterion on AB:

- Since $1_G \in A$ and $1_G \in B$, $1_G = 1_G \cdot 1_G \in AB$, so $AB \neq \emptyset$.
- Let $x, y \in AB$. Write $x = a_1b_1$ and $y = a_2b_2$ with $a_i \in A, b_i \in B$. Then:

$$\begin{aligned}xy^{-1} &= (a_1b_1)(a_2b_2)^{-1} = a_1b_1b_2^{-1}a_2^{-1} \\ &= a_1(b_1b_2^{-1})a_2^{-1}.\end{aligned}$$

  Let $b_3 = b_1b_2^{-1} \in B$. Since $B \trianglelefteq G$, $a_2^{-1}(a_2b_3a_2^{-1}) = b_3a_2^{-1}$, so $b_3a_2^{-1} = a_2^{-1}b_4$ for some $b_4 \in B$. Then:

$$xy^{-1} = a_1(a_2^{-1}b_4) = (a_1a_2^{-1})b_4 \in AB.$$

  Thus $AB \leq G$.

(ii) Proof that $B \trianglelefteq AB$: Since $B \trianglelefteq G$ and $B \subseteq AB \subseteq G$, for any $x \in AB$, $xbx^{-1} \in B$ because normality holds for all elements of G. Thus $B \trianglelefteq AB$.

(iii) Proof of Isomorphism and $A \cap B \trianglelefteq A$: Define the mapping:

$$\begin{aligned} \varphi : A &\longrightarrow AB/B \\ a &\longmapsto aB. \end{aligned}$$

- *Homomorphism:* For any $a_1, a_2 \in A$:

$$\varphi(a_1a_2) = (a_1a_2)B = (a_1B)(a_2B) = \varphi(a_1)\varphi(a_2).$$

- *Surjectivity:* Any coset in AB/B has the form $(ab)B$ for $a \in A, b \in B$. Since $b \in B$, $bB = B$. Thus $(ab)B = a(bB) = aB = \varphi(a)$. Hence φ is surjective onto AB/B.
- *Kernel Calculation:*

$$\begin{aligned} \ker(\varphi) &= \{a \in A \mid \varphi(a) = 1_{AB/B}\} \\ &= \{a \in A \mid aB = B\} \\ &= \{a \in A \mid a \in B\} \\ &= A \cap B. \end{aligned}$$

By the First Isomorphism Theorem (Theorem 2.5):

$$\ker(\varphi) = A \cap B \trianglelefteq A \quad \text{and} \quad A/(A \cap B) \cong AB/B.$$

□

Theorem 2.7 (Third Isomorphism Theorem)**.** *Let G be a group and let $H, K \trianglelefteq G$ with $H \leq K$. Then:*

(i) $K/H \trianglelefteq G/H$.

(ii) $(G/H)/(K/H) \cong G/K$.

Proof. Define the projection mapping:

$$\begin{aligned} \psi : G/H &\longrightarrow G/K \\ gH &\longmapsto gK. \end{aligned}$$

1. **Well-Definedness:** Suppose $g_1H = g_2H$. Then $g_2^{-1}g_1 \in H$. Since $H \subseteq K$, this implies $g_2^{-1}g_1 \in K$. By Lemma 1.18(iii), $g_1K = g_2K$. Thus $\psi(g_1H) = \psi(g_2H)$, so ψ is well-defined.
2. **Homomorphism Property:** For any $aH, bH \in G/H$:

$$\psi((aH)(bH)) = \psi((ab)H) = (ab)K = (aK)(bK) = \psi(aH)\psi(bH).$$

3. **Surjectivity:** For any $gK \in G/K$, the coset $gH \in G/H$ satisfies $\psi(gH) = gK$. Thus ψ is surjective.
4. **Kernel Calculation:**

$$\begin{aligned} \ker(\psi) &= \{gH \in G/H \mid \psi(gH) = 1_{G/K}\} \\ &= \{gH \in G/H \mid gK = K\} \\ &= \{gH \in G/H \mid g \in K\} \\ &= K/H. \end{aligned}$$

Applying the First Isomorphism Theorem to ψ proves:

$$K/H = \ker(\psi) \trianglelefteq G/H \quad \text{and} \quad (G/H)/(K/H) \cong G/K.$$

$\square$

Theorem 2.8 (Fourth Isomorphism Theorem / Lattice Correspondence Theorem)**.** *Let $N \trianglelefteq G$. Let $\mathcal{S}(G,N) = \{A \leq G \mid N \leq A\}$ be the collection of subgroups of G containing N, and let $\mathcal{S}(G/N)$ be the collection of all subgroups of G/N. The mapping:*

$$\begin{aligned} \Phi : \mathcal{S}(G,N) &\longrightarrow \mathcal{S}(G/N) && (2.7)\\ A &\longmapsto A/N = \{aN \mid a \in A\} && (2.8) \end{aligned}$$

is an inclusion-preserving bijection. Its two-sided inverse mapping is:

$$\begin{aligned} \Psi : \mathcal{S}(G/N) &\longrightarrow \mathcal{S}(G,N) && (2.9)\\ \bar{A} &\longmapsto \pi^{-1}(\bar{A}) = \{g \in G \mid gN \in \bar{A}\}, && (2.10) \end{aligned}$$

where $\pi : G \to G/N$ is the canonical projection.

Furthermore, this correspondence satisfies the following structural properties for all $A, B \in \mathcal{S}(G,N)$:

(i) Order/Inclusion Preservation: *$A \leq B \iff A/N \leq B/N$.*

(ii) Index Preservation: *If $A \leq B$, then $[B : A] = [B/N : A/N]$.*

(iii) Join Preservation: *$\langle A, B\rangle/N = \langle A/N, B/N\rangle$.*

(iv) Intersection Preservation: *$(A \cap B)/N = (A/N) \cap (B/N)$.*

(v) Normality Preservation: *$A \trianglelefteq G \iff A/N \trianglelefteq G/N$, and in this case:*

$$(G/N)/(A/N) \cong G/A. \tag{2.11}$$

Proof. 1. **Proof that Φ and Ψ are Mutual Inverses (Bijectivity):**

- Let $A \in \mathcal{S}(G,N)$. Since $N \trianglelefteq A$ (because $N \trianglelefteq G$), A/N is a well-defined subgroup of G/N. Now compute $\Psi(\Phi(A)) = \pi^{-1}(A/N)$:

$$\pi^{-1}(A/N) = \{g \in G \mid gN \in A/N\} = \{g \in G \mid \exists a \in A, gN = aN\}.$$

  $gN = aN \iff a^{-1}g \in N \subseteq A \iff g \in aN \subseteq A$. Thus $\pi^{-1}(A/N) = A$, so $\Psi \circ \Phi = \text{id}$.
- Let $\bar{A} \leq G/N$. First, $\pi^{-1}(\bar{A})$ is a subgroup of G containing $\ker(\pi) = N$. Now compute $\Phi(\Psi(\bar{A})) = \pi(\pi^{-1}(\bar{A}))$. Since π is surjective, $\pi(\pi^{-1}(\bar{A})) = \bar{A}$. Thus $\Phi \circ \Psi = \text{id}$.

Therefore, Φ is a bijection.

2. **Proof of (i) (Inclusion):** $A \subseteq B \implies A/N = \{aN \mid a \in A\} \subseteq \{bN \mid b \in B\} = B/N$. Conversely, $A/N \subseteq B/N \implies A = \pi^{-1}(A/N) \subseteq \pi^{-1}(B/N) = B$.

3. **Proof of (ii) (Index):** Define a map between coset spaces $\Gamma : \{bA \mid b \in B\} \to \{(bN)(A/N) \mid b \in B\}$ by $\Gamma(bA) = (bN)(A/N)$. Since $b_1A = b_2A \iff b_2^{-1}b_1 \in A \iff (b_2^{-1}b_1)N \in A/N \iff (b_2N)^{-1}(b_1N) \in A/N \iff (b_1N)(A/N) = (b_2N)(A/N)$, Γ is a well-defined bijection. Thus $[B : A] = [B/N : A/N]$.

4. **Proof of (iv) (Intersection):** $xN \in (A \cap B)/N \iff x \in A \cap B \iff x \in A$ and $x \in B \iff xN \in A/N$ and $xN \in B/N \iff xN \in (A/N) \cap (B/N)$.

5. **Proof of (iii) (Join):** Since $A \leq \langle A, B\rangle$ and $B \leq \langle A, B\rangle$, by (i) we have $A/N \leq \langle A, B\rangle/N$ and $B/N \leq \langle A, B\rangle/N$, so $\langle A/N, B/N\rangle \leq \langle A, B\rangle/N$. Conversely, $\pi^{-1}(\langle A/N, B/N\rangle)$ is a subgroup of G containing both $\pi^{-1}(A/N) = A$ and $\pi^{-1}(B/N) = B$, so it contains $\langle A, B\rangle$. Applying π gives $\langle A, B\rangle/N \leq \langle A/N, B/N\rangle$.

6. **Proof of (v) (Normality):**

$$\begin{aligned} A \trianglelefteq G &\iff \forall g \in G,\ gAg^{-1} = A \\ &\iff \forall g \in G,\ (gN)(A/N)(gN)^{-1} = A/N \\ &\iff A/N \trianglelefteq G/N. \end{aligned}$$

When $A \trianglelefteq G$, the isomorphism $(G/N)/(A/N) \cong G/A$ is precisely the Third Isomorphism Theorem (Theorem 2.7).

□

2.3 Direct Products of Groups

Definition 2.9 (External Direct Product)**.** Let $G_1, G_2, \ldots, G_n$ be groups. The **external direct product**:

$$G_1 \times G_2 \times \cdots \times G_n = \{(g_1, g_2, \ldots, g_n) \mid g_i \in G_i\} \tag{2.12}$$

is a group under componentwise multiplication:

$$(g_1, g_2, \ldots, g_n)(h_1, h_2, \ldots, h_n) = (g_1h_1, g_2h_2, \ldots, g_nh_n). \tag{2.13}$$

The identity is $(1_{G_1}, \ldots, 1_{G_n})$ and $(g_1, \ldots, g_n)^{-1} = (g_1^{-1}, \ldots, g_n^{-1})$.

Proposition 2.10 (Properties of Direct Products)**.** *Let $G = G_1 \times G_2 \times \cdots \times G_n$.*

(i) $|G| = |G_1| \cdot |G_2| \cdots |G_n|$.

(ii) *The order of an element $(g_1, \ldots, g_n) \in G$ is:*

$$|(g_1, \ldots, g_n)| = \operatorname{lcm}(|g_1|, \ldots, |g_n|). \tag{2.14}$$

(iii) *G is abelian if and only if each G_i is abelian.*

(iv) $\mathbb{Z}_m \times \mathbb{Z}_n \cong \mathbb{Z}_{mn} \iff \gcd(m, n) = 1$.

Proof. **(i)** Direct consequence of the rule of product for Cartesian sets.

(ii) For any integer $k \geq 1$:

$$\begin{aligned} (g_1, \ldots, g_n)^k = (1, \ldots, 1) &\iff (g_1^k, \ldots, g_n^k) = (1, \ldots, 1) \\ &\iff g_i^k = 1_{G_i} \text{ for all } i \in \{1, \ldots, n\} \\ &\iff |g_i| \text{ divides } k \text{ for all } i \\ &\iff \operatorname{lcm}(|g_1|, \ldots, |g_n|) \text{ divides } k. \end{aligned}$$

Thus the smallest positive integer is $k = \operatorname{lcm}(|g_1|, \ldots, |g_n|)$.

(iii) Commutativity holds componentwise: $(g_1h_1, \ldots) = (h_1g_1, \ldots) \iff g_ih_i = h_ig_i$ for all i.

(iv) Consider the generator candidate $(1, 1) \in \mathbb{Z}_m \times \mathbb{Z}_n$. Its order is:

$$|(1, 1)| = \mathrm{lcm}(m, n) = \frac{mn}{\gcd(m, n)}.$$

Since $|\mathbb{Z}_m \times \mathbb{Z}_n| = mn$, the group has an element of order mn if and only if $\mathrm{lcm}(m, n) = mn$, which occurs if and only if $\gcd(m, n) = 1$.

□

Theorem 2.11 (Internal Direct Product Recognition)**.** *Let G be a group with subgroups H and K. Then $G \cong H \times K$ if and only if:*

(1) $H \trianglelefteq G$ *and* $K \trianglelefteq G$*,*

(2) $H \cap K = \{1_G\}$*, and*

(3) $HK = G$*.*

Proof. • **Direction ($\implies$):** If $G \cong H \times K$, identifying H with $H \times \{1_K\}$ and K with $\{1_H\} \times K$ makes both subgroups normal, with intersection $(H \times \{1\}) \cap (\{1\} \times K) = \{(1, 1)\}$ and product G.

- **Direction ($\impliedby$):** First, we show that elements of H and K commute. Let $h \in H$ and $k \in K$. Consider the commutator:

$$[h, k] = hkh^{-1}k^{-1} = (hkh^{-1})k^{-1} = h(kh^{-1}k^{-1}).$$

  – Since $K \trianglelefteq G$, $hkh^{-1} \in K$, so $[h, k] = (hkh^{-1})k^{-1} \in K$.
  – Since $H \trianglelefteq G$, $kh^{-1}k^{-1} \in H$, so $[h, k] = h(kh^{-1}k^{-1}) \in H$.

  Therefore, $[h, k] \in H \cap K$. Since $H \cap K = \{1_G\}$:

$$hkh^{-1}k^{-1} = 1_G \implies hk = kh.$$

  Now define the multiplication map:

$$\begin{aligned} \theta : H \times K &\longrightarrow G \\ (h, k) &\longmapsto hk. \end{aligned}$$

  – *Homomorphism:* Using $k_1h_2 = h_2k_1$:

$$\begin{aligned} \theta((h_1, k_1)(h_2, k_2)) &= \theta(h_1h_2, k_1k_2) \\ &= (h_1h_2)(k_1k_2) \\ &= h_1(h_2k_1)k_2 \\ &= h_1(k_1h_2)k_2 \\ &= (h_1k_1)(h_2k_2) \\ &= \theta(h_1, k_1)\theta(h_2, k_2). \end{aligned}$$

  – *Injectivity:* Suppose $\theta(h, k) = 1_G$. Then:

$$hk = 1_G \implies h = k^{-1} \in H \cap K = \{1_G\} \implies h = 1_G \text{ and } k = 1_G.$$

  Thus $\ker(\theta) = \{(1_G, 1_G)\}$, so θ is injective.

  – *Surjectivity:* Since $HK = G$, every $g \in G$ can be written as $g = hk = \theta(h, k)$ for some $h \in H, k \in K$.

  Thus θ is an isomorphism, so $G \cong H \times K$.

□

2.4 Commutators and Abelianization

Definition 2.12 (Commutator)**.** For $x, y \in G$, the **commutator** of x and y is:

$$[x, y] = x^{-1}y^{-1}xy. \tag{2.15}$$

Notice that $[x, y] = 1_G \iff xy = yx$.

Definition 2.13 (Commutator Subgroup / Derived Subgroup)**.** The **commutator subgroup** of G, denoted G' or $[G, G]$, is the subgroup generated by all commutators:

$$G' = [G, G] = \langle [x, y] \mid x, y \in G \rangle. \tag{2.16}$$

Theorem 2.14 (Properties of the Commutator Subgroup and Abelianization)**.** *Let G be a group.*

(i) *$G' \trianglelefteq G$.*

(ii) *The quotient group G/G' is abelian, called the **abelianization** of G, denoted G^{ab}.*

(iii) ***Universal Property of Abelianization**: If $N \trianglelefteq G$, then G/N is abelian if and only if $G' \leq N$.*

(iv) *Any homomorphism $\varphi : G \to A$ into an abelian group A factors uniquely through $G^{ab} = G/G'$.*

Proof. **(i) Normality of G':** Let $g \in G$ and let $[x, y] \in G'$. Compute the conjugate of the commutator:

$$\begin{aligned} g[x, y]g^{-1} &= g(x^{-1}y^{-1}xy)g^{-1} \\ &= (gx^{-1}g^{-1})(gy^{-1}g^{-1})(gxg^{-1})(gyg^{-1}) \\ &= (gxg^{-1})^{-1}(gyg^{-1})^{-1}(gxg^{-1})(gyg^{-1}) \\ &= [gxg^{-1}, gyg^{-1}] \in G'. \end{aligned}$$

Since the conjugate of every generator of G' lies in G', $gG'g^{-1} = G'$ for all $g \in G$. Thus $G' \trianglelefteq G$.

(ii) G/G' is Abelian: Let $xG', yG' \in G/G'$. Then:

$$\begin{aligned} (xG')(yG') &= (xy)G' \\ &= (yx(x^{-1}y^{-1}xy))G' \\ &= (yx[x, y])G' \\ &= (yx)G' \quad \text{(since } [x, y] \in G'\text{)} \\ &= (yG')(xG'). \end{aligned}$$

Thus G/G' is abelian.

(iii) Universal Criterion for Abelian Quotients: For any $N \trianglelefteq G$:

$$\begin{aligned} G/N \text{ is abelian} &\iff \forall x, y \in G,\ (xN)(yN) = (yN)(xN) \\ &\iff \forall x, y \in G,\ (xy)N = (yx)N \\ &\iff \forall x, y \in G,\ (yx)^{-1}(xy) \in N \\ &\iff \forall x, y \in G,\ x^{-1}y^{-1}xy \in N \\ &\iff \forall x, y \in G,\ [x, y] \in N \\ &\iff G' = \langle [x, y] \rangle \leq N. \end{aligned}$$

(iv) Factorization of Homomorphisms: Let $\varphi : G \to A$ where A is an abelian group. By the First Isomorphism Theorem, $G/\ker(\varphi) \cong \mathrm{im}(\varphi) \leq A$. Since A is abelian, $\mathrm{im}(\varphi)$ is abelian, so $G/\ker(\varphi)$ is abelian. By part (iii), $G' \leq \ker(\varphi)$.

Define $\bar{\varphi} : G/G' \to A$ by $\bar{\varphi}(gG') = \varphi(g)$. If $g_1G' = g_2G'$, then $g_2^{-1}g_1 \in G' \leq \ker(\varphi)$, so:

$$\varphi(g_2^{-1}g_1) = 1_A \implies \varphi(g_1) = \varphi(g_2).$$

Thus $\bar{\varphi}$ is well-defined, and it is the unique homomorphism making the diagram commute. □

2.5 Exercises and Step-by-Step Solutions

Problem 2.15 (Exercise 1: Image of a Normal Subgroup)**.** Let $\varphi : G \to H$ be a **surjective** homomorphism of groups. Prove that if $N \trianglelefteq G$, then $\varphi(N) \trianglelefteq H$.

Solution. First, $\varphi(N)$ is a subgroup of H by Proposition 1.5(ii). To prove normality, let $h \in H$ and let $y \in \varphi(N)$. By definition of $\varphi(N)$, $y = \varphi(n)$ for some $n \in N$. Since φ is surjective, there exists $g \in G$ such that $\varphi(g) = h$. Now compute the conjugate hyh^{-1} in H:

$$\begin{aligned} hyh^{-1} &= \varphi(g)\varphi(n)\big(\varphi(g)\big)^{-1} \\ &= \varphi(g)\varphi(n)\varphi(g^{-1}) \quad \text{(by Proposition 1.2(ii))} \\ &= \varphi(gng^{-1}) \quad \text{(homomorphism property).} \end{aligned}$$

Since $N \trianglelefteq G$, $gng^{-1} \in N$. Therefore, $\varphi(gng^{-1}) \in \varphi(N)$, so $hyh^{-1} \in \varphi(N)$. Thus $\varphi(N) \trianglelefteq H$. □

Problem 2.16 (Exercise 2: Intersection with Normal Subgroup)**.** Let $H \leq G$ and $N \trianglelefteq G$. Prove that $H \cap N \trianglelefteq H$.

Solution. By Proposition 1.14, $H \cap N \leq G$, and since $H \cap N \subseteq H$, we have $H \cap N \leq H$. Let $h \in H$ and let $x \in H \cap N$. We must show $hxh^{-1} \in H \cap N$:

- Since $h \in H$ and $x \in H$, and H is a subgroup, $hxh^{-1} \in H$.
- Since $h \in G$ and $x \in N$, and $N \trianglelefteq G$, $hxh^{-1} \in N$.

Therefore, $hxh^{-1} \in H \cap N$ for all $h \in H$ and $x \in H \cap N$. Thus $H \cap N \trianglelefteq H$. □

Problem 2.17 (Exercise 3: Cyclic Quotient by Center)**.** Show that if $G/Z(G)$ is cyclic, then G is abelian (and hence $G = Z(G)$ and $G/Z(G) \cong \{1\}$).

Solution. Suppose $G/Z(G)$ is cyclic. Let $gZ(G)$ be a generator of $G/Z(G)$, so:

$$G/Z(G) = \langle gZ(G)\rangle = \{g^kZ(G) \mid k \in \mathbb{Z}\}.$$

Let $x, y \in G$ be arbitrary elements. Since cosets partition G, $x \in g^aZ(G)$ and $y \in g^bZ(G)$ for some integers $a, b \in \mathbb{Z}$. Thus there exist $z_1, z_2 \in Z(G)$ such that:

$$x = g^az_1 \quad \text{and} \quad y = g^bz_2.$$

Now compute the product xy:

$$\begin{aligned}
xy &= (g^a z_1)(g^b z_2) \\
&= g^a (z_1 g^b) z_2 \\
&= g^a (g^b z_1) z_2 \quad \text{(since } z_1 \in Z(G) \text{ commutes with all elements)} \\
&= (g^a g^b)(z_1 z_2) \\
&= g^{a+b}(z_2 z_1) \quad \text{(since } Z(G) \text{ is abelian)} \\
&= (g^b g^a)(z_2 z_1) \\
&= g^b (g^a z_2) z_1 \\
&= g^b (z_2 g^a) z_1 \quad \text{(since } z_2 \in Z(G)) \\
&= (g^b z_2)(g^a z_1) \\
&= yx.
\end{aligned}$$

Since $xy = yx$ for all $x, y \in G$, G is abelian. □

Problem 2.18 (Exercise 4: Intersection of Normal Subgroups)**.** Prove that the intersection of any arbitrary collection of normal subgroups of G is a normal subgroup of G.

Solution. Let $\{N_i\}_{i \in I}$ be a family of normal subgroups of G, and let $N = \bigcap_{i \in I} N_i$. By Proposition 1.14, $N \leq G$. To prove normality, let $g \in G$ and $x \in N$. Since $x \in N$, $x \in N_i$ for every $i \in I$. Since each $N_i \trianglelefteq G$, $gxg^{-1} \in N_i$ for every $i \in I$. Therefore:

$$gxg^{-1} \in \bigcap_{i \in I} N_i = N.$$

Thus $gNg^{-1} \subseteq N$ for all $g \in G$, proving $N \trianglelefteq G$. □

Problem 2.19 (Exercise 5: Commutators in S_3)**.** Let $G = S_3$. Compute the commutator subgroup G' and the abelianization G/G'.

Solution. The symmetric group S_3 has order $|S_3| = 6$:

$$S_3 = \{1, (1\ 2), (2\ 3), (1\ 3), (1\ 2\ 3), (1\ 3\ 2)\}.$$

Compute the commutator of two transpositions $(1\ 2)$ and $(2\ 3)$:

$$\begin{aligned}
[(1\ 2), (2\ 3)] &= (1\ 2)^{-1}(2\ 3)^{-1}(1\ 2)(2\ 3) \\
&= (1\ 2)(2\ 3)(1\ 2)(2\ 3) \\
&= (1\ 2\ 3)(1\ 2\ 3) \\
&= (1\ 3\ 2).
\end{aligned}$$

Its inverse is $(1\ 3\ 2)^{-1} = (1\ 2\ 3) \in G'$. Thus $A_3 = \{1, (1\ 2\ 3), (1\ 3\ 2)\} \leq G'$.

Since S_3 is non-abelian, $G' \neq \{1\}$, so $|G'| \geq 3$. Moreover, S_3/A_3 has order $|S_3|/|A_3| = 6/3 = 2$, which is cyclic and therefore abelian. By Theorem 2.14(iii), $G' \leq A_3$. Since $A_3 \leq G'$ and $G' \leq A_3$, we conclude:

$$G' = [S_3, S_3] = A_3 \cong \mathbb{Z}_3.$$

The abelianization is:

$$S_3^{\mathrm{ab}} = S_3/G' = S_3/A_3 \cong \mathbb{Z}_2.$$

□

Problem 2.20 (Exercise 6: Groups of Exponent 2)**.** Let G be a group such that $x^2 = 1_G$ for all $x \in G$. Prove that G is abelian and that $G' = \{1_G\}$.

Solution. Since $x^2 = 1_G$ for all $x \in G$, multiplying by x^{-1} gives $x = x^{-1}$ for all $x \in G$. Now let $a, b \in G$. Since $ab \in G$, we also have $(ab) = (ab)^{-1}$. Using the socks-shoes property for group inverses:

$$\begin{aligned} ab &= (ab)^{-1} \\ &= b^{-1}a^{-1} \\ &= ba \quad (\text{since } b^{-1} = b \text{ and } a^{-1} = a). \end{aligned}$$

Thus $ab = ba$ for all $a, b \in G$, so G is abelian.

Since G is abelian, for all $x, y \in G$:

$$[x, y] = x^{-1}y^{-1}xy = x^{-1}xy^{-1}y = 1_G \cdot 1_G = 1_G.$$

Therefore, the commutator subgroup is trivial: $G' = [G, G] = \{1_G\}$. □

Chapter 3

Symmetric and Alternating Groups

In this chapter, we study permutation groups: the symmetric group S_n, cycle decompositions, transpositions, the sign homomorphism, the alternating group A_n, the simplicity of A_n for $n \geq 5$, and the complete classification of all normal subgroups of S_n. Every lemma and theorem is accompanied by a complete, step-by-step mathematical proof.

3.1 The Symmetric Group and Cycle Decomposition

Definition 3.1 (Symmetric Group)**.** Let $\Omega = \{1, 2, \ldots, n\}$. The **symmetric group on n letters**, denoted S_n, is the group of all bijections $\sigma : \Omega \to \Omega$ under function composition:

$$|S_n| = n! \tag{3.1}$$

Definition 3.2 (Cycle Notation)**.** A permutation $\sigma \in S_n$ is called a **k-cycle** (or cycle of length k), denoted:

$$\sigma = (a_1\ a_2\ \ldots\ a_k), \tag{3.2}$$

if $\sigma(a_1) = a_2$, $\sigma(a_2) = a_3$, $\ldots$, $\sigma(a_{k-1}) = a_k$, $\sigma(a_k) = a_1$, and $\sigma(x) = x$ for all $x \notin \{a_1, \ldots, a_k\}$. The set $\{a_1, \ldots, a_k\}$ is called the **support** of σ, denoted $\mathrm{supp}(\sigma)$. Two cycles σ and τ are called **disjoint** if $\mathrm{supp}(\sigma) \cap \mathrm{supp}(\tau) = \emptyset$.

Proposition 3.3 (Commutativity of Disjoint Cycles)**.** *Disjoint cycles commute: if σ and τ are disjoint cycles in S_n, then:*

$$\sigma\tau = \tau\sigma. \tag{3.3}$$

Proof. Let $\sigma = (a_1\ a_2\ \ldots\ a_k)$ and $\tau = (b_1\ b_2\ \ldots\ b_m)$ with $\{a_1, \ldots, a_k\} \cap \{b_1, \ldots, b_m\} = \emptyset$. To show that the permutations $\sigma\tau$ and $\tau\sigma$ are identical functions on $\Omega = \{1, 2, \ldots, n\}$, we evaluate both compositions on any arbitrary element $x \in \Omega$:

- **Case 1: $x \in \mathrm{supp}(\sigma)$ (so $x = a_i$ for some $1 \leq i \leq k$).** Since $\mathrm{supp}(\sigma) \cap \mathrm{supp}(\tau) = \emptyset$, $a_i \notin \mathrm{supp}(\tau)$, so $\tau(a_i) = a_i$. Moreover, $\sigma(a_i) = a_{i+1} \in \mathrm{supp}(\sigma)$ (with $a_{k+1} = a_1$), so $a_{i+1} \notin \mathrm{supp}(\tau) \implies \tau(a_{i+1}) = a_{i+1}$. Computing both compositions:

$$(\sigma\tau)(a_i) = \sigma\big(\tau(a_i)\big) = \sigma(a_i) = a_{i+1},$$
$$(\tau\sigma)(a_i) = \tau\big(\sigma(a_i)\big) = \tau(a_{i+1}) = a_{i+1}.$$

  Thus $(\sigma\tau)(a_i) = (\tau\sigma)(a_i)$.

- **Case 2: $x \in \mathrm{supp}(\tau)$ (so $x = b_j$ for some $1 \leq j \leq m$).** Since $b_j \notin \mathrm{supp}(\sigma)$, $\sigma(b_j) = b_j$. Also, $\tau(b_j) = b_{j+1} \notin \mathrm{supp}(\sigma) \implies \sigma(b_{j+1}) = b_{j+1}$. Computing both compositions:

$$(\sigma\tau)(b_j) = \sigma\big(\tau(b_j)\big) = \sigma(b_{j+1}) = b_{j+1},$$
$$(\tau\sigma)(b_j) = \tau\big(\sigma(b_j)\big) = \tau(b_j) = b_{j+1}.$$

  Thus $(\sigma\tau)(b_j) = (\tau\sigma)(b_j)$.

- **Case 3:** $x \notin \text{supp}(\sigma) \cup \text{supp}(\tau)$ **(fixed by both cycles).** Then $\sigma(x) = x$ and $\tau(x) = x$. Thus:

$$(\sigma\tau)(x) = \sigma\big(\tau(x)\big) = \sigma(x) = x,$$
$$(\tau\sigma)(x) = \tau\big(\sigma(x)\big) = \tau(x) = x.$$

  Thus $(\sigma\tau)(x) = (\tau\sigma)(x)$.

In all cases, $(\sigma\tau)(x) = (\tau\sigma)(x)$ for every $x \in \Omega$. Therefore, $\sigma\tau = \tau\sigma$. □

Theorem 3.4 (Disjoint Cycle Decomposition Theorem)**.** *Every permutation $\sigma \in S_n$ can be expressed as a product of disjoint cycles. This decomposition is unique up to the order in which the cycles are written.*

Proof. Consider the action of the cyclic group $\langle\sigma\rangle$ on the set $\Omega = \{1, 2, \dots, n\}$ defined by $k \cdot x = \sigma^k(x)$. Define the relation $\sim$ on Ω by:

$$x \sim y \iff \exists m \in \mathbb{Z} \text{ such that } \sigma^m(x) = y.$$

1. **$\sim$ is an Equivalence Relation:**
   - *Reflexive:* $\sigma^0(x) = x \implies x \sim x$.
   - *Symmetric:* $y = \sigma^m(x) \implies x = \sigma^{-m}(y) \implies y \sim x$.
   - *Transitive:* $y = \sigma^m(x)$ and $z = \sigma^\ell(y) \implies z = \sigma^{m+\ell}(x) \implies x \sim z$.
2. **Cycle Structure of Orbits:** The equivalence classes under $\sim$ partition Ω into disjoint orbits $\mathcal{O}_1, \mathcal{O}_2, \dots, \mathcal{O}_r$. For any orbit $\mathcal{O}$, choose an element $x \in \mathcal{O}$. Consider the sequence $x, \sigma(x), \sigma^2(x), \dots$. Since Ω is finite, there must be a smallest positive integer $k \geq 1$ such that $\sigma^k(x) = x$. The elements:
$$x,\ \sigma(x),\ \sigma^2(x),\ \dots,\ \sigma^{k-1}(x)$$
   are all distinct and constitute the entire orbit $\mathcal{O}$. The permutation σ acts on $\mathcal{O}$ precisely as the k-cycle:
$$c = (x\ \sigma(x)\ \sigma^2(x)\ \dots\ \sigma^{k-1}(x)).$$
3. **Product of Cycles:** Since the orbits $\mathcal{O}_i$ are mutually disjoint and partition Ω, the corresponding cycles c_i have disjoint supports. For each $x \in \Omega$, x belongs to a unique orbit $\mathcal{O}_j$, so:
$$(c_1 c_2 \cdots c_r)(x) = c_j(x) = \sigma(x).$$
   Omitting 1-cycles (which correspond to fixed points and act as the identity permutation) produces the disjoint cycle decomposition:
$$\sigma = c_1 c_2 \cdots c_m.$$
4. **Uniqueness:** The cycles in the decomposition are uniquely determined by the partition of Ω into orbits of $\langle\sigma\rangle$ and the cyclic permutation of elements within each orbit. Since disjoint cycles commute (Proposition 3.3), the factorization is unique up to the ordering of the cycles.

□

Proposition 3.5 (Order of a Permutation)**.** *If $\sigma \in S_n$ has disjoint cycle decomposition $\sigma = c_1 c_2 \cdots c_r$, where c_i is a cycle of length L_i, then the order of σ is:*

$$|\sigma| = \text{lcm}(L_1, L_2, \dots, L_r). \tag{3.4}$$

Proof. Since disjoint cycles commute (Proposition 3.3):

$$\sigma^m = (c_1 c_2 \cdots c_r)^m = c_1^m c_2^m \cdots c_r^m.$$

Since the cycles c_i have pairwise disjoint supports:

$$\begin{aligned} \sigma^m = \mathrm{id} &\iff c_1^m c_2^m \cdots c_r^m = \mathrm{id} \\ &\iff c_i^m = \mathrm{id} \quad \text{for all } i \in \{1, 2, \dots, r\} \\ &\iff L_i \mid m \quad \text{for all } i \in \{1, 2, \dots, r\} \quad (\text{since the order of a } k\text{-cycle is } k) \\ &\iff \mathrm{lcm}(L_1, L_2, \dots, L_r) \mid m. \end{aligned}$$

The smallest positive integer m satisfying this condition is $m = \mathrm{lcm}(L_1, L_2, \dots, L_r)$. □

3.2 Transpositions, Parity, and the Alternating Group

Definition 3.6 (Transposition)**.** A 2-cycle $(i\ j) \in S_n$ (with $i \neq j$) is called a **transposition**.

Proposition 3.7 (Generation of S_n by Transpositions)**.** *Every permutation in S_n ($n \geq 2$) can be written as a product of transpositions.*

Proof. By Theorem 3.4, every permutation is a product of disjoint cycles. It therefore suffices to show that any k-cycle $(a_1\ a_2\ \dots\ a_k)$ can be factored into transpositions. We verify by direct algebraic expansion:

$$(a_1\ a_2\ \dots\ a_k) = (a_1\ a_k)(a_1\ a_{k-1}) \cdots (a_1\ a_3)(a_1\ a_2). \tag{3.5}$$

Evaluating the right-hand composition on each symbol a_i:

- $a_1 \mapsto a_2$ via $(a_1\ a_2)$, which is fixed by all subsequent transpositions.
- a_i $(2 \leq i \leq k-1)$: $(a_1\ a_i)$ maps $a_i \mapsto a_1$, and the next transposition $(a_1\ a_{i+1})$ maps $a_1 \mapsto a_{i+1}$, which is then fixed by the remaining transpositions. Thus $a_i \mapsto a_{i+1}$.
- a_k: $(a_1\ a_k)$ maps $a_k \mapsto a_1$.

Thus $(a_1\ a_k) \cdots (a_1\ a_2) = (a_1\ a_2\ \dots\ a_k)$. Since every permutation is a product of cycles, and every cycle is a product of transpositions, every permutation is a product of transpositions. □

Definition 3.8 (Sign / Parity Homomorphism)**.** Let $x_1, x_2, \dots, x_n$ be independent variables. The **discriminant polynomial** $\Delta \in \mathbb{Z}[x_1, \dots, x_n]$ is:

$$\Delta = \prod_{1 \leq i < j \leq n} (x_j - x_i). \tag{3.6}$$

For any $\sigma \in S_n$, define the action of σ on polynomials by $\sigma(f(x_1, \dots, x_n)) = f(x_{\sigma(1)}, \dots, x_{\sigma(n)})$. The **sign** (or **signature**) of σ, denoted $\mathrm{sgn}(\sigma)$, is defined by:

$$\sigma(\Delta) = \prod_{1 \leq i < j \leq n} (x_{\sigma(j)} - x_{\sigma(i)}) = \mathrm{sgn}(\sigma) \cdot \Delta. \tag{3.7}$$

A permutation σ is called **even** if $\mathrm{sgn}(\sigma) = +1$, and **odd** if $\mathrm{sgn}(\sigma) = -1$.

Theorem 3.9 (Fundamental Properties of the Sign Homomorphism)**.** ***(i)*** *$\mathrm{sgn} : S_n \to \{+1, -1\}$ is a group homomorphism:*

$$\mathrm{sgn}(\sigma\tau) = \mathrm{sgn}(\sigma)\,\mathrm{sgn}(\tau). \tag{3.8}$$

(ii) *For any transposition $(i\ j)$, $\mathrm{sgn}((i\ j)) = -1$.*

(iii) *If σ is written as a product of k transpositions $\sigma = \tau_1\tau_2\cdots\tau_k$, then:*

$$\operatorname{sgn}(\sigma) = (-1)^k. \tag{3.9}$$

In particular, the parity of the number of transpositions in any factorization of σ is invariant.

(iv) *A k-cycle has sign $(-1)^{k-1}$.*

Proof. **(i) Homomorphism Property:** For any $\sigma, \tau \in S_n$:

$$\begin{aligned}
\operatorname{sgn}(\sigma\tau)\cdot\Delta = (\sigma\tau)(\Delta) &= \sigma\big(\tau(\Delta)\big)\\
&= \sigma\big(\operatorname{sgn}(\tau)\cdot\Delta\big)\\
&= \operatorname{sgn}(\tau)\cdot\sigma(\Delta)\\
&= \operatorname{sgn}(\tau)\cdot\big(\operatorname{sgn}(\sigma)\cdot\Delta\big)\\
&= \big(\operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)\big)\cdot\Delta.
\end{aligned}$$

Cancelling $\Delta \neq 0$ gives $\operatorname{sgn}(\sigma\tau) = \operatorname{sgn}(\sigma)\operatorname{sgn}(\tau)$.

(ii) Sign of a Transposition: Consider first an adjacent transposition $\tau = (k\ k+1)$. In the product $\Delta = \prod_{i<j}(x_j - x_i)$:

- The single factor $(x_{k+1} - x_k)$ is changed to $(x_k - x_{k+1}) = -(x_{k+1} - x_k)$, contributing a sign change.
- For any $i < k$, the product of the two factors $(x_k - x_i)(x_{k+1} - x_i)$ is mapped to $(x_{k+1} - x_i)(x_k - x_i)$, which is unchanged.
- For any $j > k+1$, the product $(x_j - x_k)(x_j - x_{k+1})$ is mapped to $(x_j - x_{k+1})(x_j - x_k)$, which is unchanged.
- All other factors do not involve k or $k+1$ and are completely unchanged.

Therefore, $\tau(\Delta) = -\Delta$, so $\operatorname{sgn}\big((k\ k+1)\big) = -1$.

Now consider any arbitrary transposition $(i\ j)$ with $i < j$. We can write $(i\ j)$ as a conjugate of the adjacent transposition $(i\ i+1)$ using the permutation $\rho = (i+1\ j)$:

$$(i\ j) = \rho(i\ i+1)\rho^{-1}.$$

Applying the homomorphism property:

$$\operatorname{sgn}\big((i\ j)\big) = \operatorname{sgn}(\rho)\cdot\operatorname{sgn}\big((i\ i+1)\big)\cdot\operatorname{sgn}(\rho^{-1}) = \operatorname{sgn}(\rho)\cdot(-1)\cdot\big(\operatorname{sgn}(\rho)\big)^{-1} = -1.$$

(iii) Invariance of Transposition Parity: If $\sigma = \tau_1\tau_2\cdots\tau_k$ where each τ_i is a transposition:

$$\operatorname{sgn}(\sigma) = \operatorname{sgn}(\tau_1)\operatorname{sgn}(\tau_2)\cdots\operatorname{sgn}(\tau_k) = (-1)(-1)\cdots(-1) = (-1)^k.$$

Since $\operatorname{sgn}(\sigma)$ is intrinsically defined via $\sigma(\Delta)$, the value $(-1)^k$ is unique. Thus $k \pmod 2$ is independent of the chosen factorization.

(iv) Sign of a Cycle: By Proposition 3.7, a k-cycle factors into $k-1$ transpositions:

$$\operatorname{sgn}\big((a_1\ a_2\ \ldots\ a_k)\big) = (-1)^{k-1}.$$

□

Definition 3.10 (Alternating Group)**.** The **alternating group on n letters**, denoted A_n, is the group of all even permutations in S_n:

$$A_n = \ker(\mathrm{sgn}) = \{\sigma \in S_n \mid \mathrm{sgn}(\sigma) = +1\}. \tag{3.10}$$

Proposition 3.11 (Structural Properties of A_n)**.** *For any $n \geq 2$:*

(i) *$A_n \trianglelefteq S_n$ and $[S_n : A_n] = 2$.*

(ii) *$|A_n| = \frac{n!}{2}$.*

(iii) *A_n is generated by the set of all 3-cycles.*

Proof. **(i)** Since $A_n = \ker(\mathrm{sgn})$ and $\mathrm{sgn} : S_n \to \{+1, -1\}$ is a homomorphism into the group of order 2, Proposition 1.5(i) implies $A_n \trianglelefteq S_n$. By the First Isomorphism Theorem:

$$S_n/A_n \cong \mathrm{im}(\mathrm{sgn}) = \{+1, -1\} \implies [S_n : A_n] = |\{+1, -1\}| = 2.$$

(ii) By Lagrange's Theorem:

$$|A_n| = \frac{|S_n|}{[S_n : A_n]} = \frac{n!}{2}.$$

(iii) Every element of A_n is a product of an even number of transpositions. We can group these transpositions into consecutive pairs $\tau_1\tau_2$. There are three cases for any pair of transpositions $(a\ b)(c\ d)$:

- *Case 1: $\{a, b\} = \{c, d\}$ (identical transpositions).*

$$(a\ b)(a\ b) = \mathrm{id} = (a\ b\ c)^3 \quad (\text{for any } c \notin \{a, b\}).$$

- *Case 2: $|\{a, b\} \cap \{c, d\}| = 1$ (overlapping transpositions).*

$$(a\ b)(b\ c) = (a\ b\ c).$$

- *Case 3: $\{a, b\} \cap \{c, d\} = \emptyset$ (disjoint transpositions).*

$$(a\ b)(c\ d) = (a\ b)(b\ c)(b\ c)(c\ d) = (a\ b\ c)(b\ c\ d).$$

In every case, a pair of transpositions is equal to a product of 3-cycles. Therefore, the set of all 3-cycles generates A_n.

□

3.3 Normal Subgroups of S_n and Simplicity of A_n

Definition 3.12 (Simple Group)**.** A group G is called **simple** if $|G| > 1$ and its only normal subgroups are the trivial subgroup $\{1_G\}$ and G itself.

Lemma 3.13 (Conjugacy of 3-Cycles in A_n)**.** *For all $n \geq 5$, all 3-cycles are conjugate to one another in A_n.*

Proof. Let $(a_1\ a_2\ a_3)$ and $(b_1\ b_2\ b_3)$ be any two 3-cycles in A_n. In S_n, choose any permutation $\rho \in S_n$ such that $\rho(a_i) = b_i$ for $i \in \{1, 2, 3\}$. Then $\rho(a_1\ a_2\ a_3)\rho^{-1} = (b_1\ b_2\ b_3)$.

- If $\mathrm{sgn}(\rho) = +1$, then $\rho \in A_n$, so the two 3-cycles are conjugate in A_n.

- If $\operatorname{sgn}(\rho) = -1$, since $n \geq 5$, there exist at least two distinct elements $u, v \in \Omega \setminus \{b_1, b_2, b_3\}$. Consider the transposition $(u\ v) \in S_n$, and define $\rho' = (u\ v)\rho$. Then:

$$\operatorname{sgn}(\rho') = \operatorname{sgn}(u\ v)\operatorname{sgn}(\rho) = (-1)(-1) = +1 \implies \rho' \in A_n.$$

  Since $u, v \notin \{b_1, b_2, b_3\}$, the transposition $(u\ v)$ commutes with $(b_1\ b_2\ b_3)$. Therefore:

$$\begin{aligned}\rho'(a_1\ a_2\ a_3)(\rho')^{-1} &= (u\ v)\rho(a_1\ a_2\ a_3)\rho^{-1}(u\ v)\\ &= (u\ v)(b_1\ b_2\ b_3)(u\ v)\\ &= (b_1\ b_2\ b_3).\end{aligned}$$

Thus in all cases, any two 3-cycles are conjugate in A_n. □

Theorem 3.14 (Simplicity of A_n for $n \geq 5$)**.** *The alternating group A_n is simple for all $n \geq 5$.*

Proof. Let N be a non-trivial normal subgroup of A_n ($N \trianglelefteq A_n$ with $N \neq \{1\}$). We will show that N must contain at least one 3-cycle. Since $N \neq \{1\}$, choose $\sigma \in N \setminus \{1\}$. Consider the disjoint cycle decomposition of σ:

- **Case 1: σ contains a cycle of length $k \geq 4$.** Write $\sigma = (1\ 2\ 3\ 4 \ldots c_k)\mu$ where μ is disjoint from $(1\ \ldots\ c_k)$. Let $\tau = (1\ 2\ 3) \in A_n$. Since $N \trianglelefteq A_n$, the conjugate $\sigma_1 = \tau\sigma\tau^{-1} \in N$, and thus the commutator:

$$\sigma' = \sigma_1\sigma^{-1} = (\tau\sigma\tau^{-1})\sigma^{-1} \in N.$$

  Computing $\sigma_1 = (1\ 2\ 3)(1\ 2\ 3\ 4 \ldots)\mu(1\ 3\ 2) = (2\ 3\ 1\ 4 \ldots)\mu = (1\ 4 \ldots 2\ 3)\mu$. Then:

$$\sigma' = \sigma_1\sigma^{-1} = (1\ 4 \ldots 2\ 3)\mu \cdot \mu^{-1}(c_k \ldots 4\ 3\ 2\ 1) = (1\ 2\ 4).$$

  Thus N contains the 3-cycle $(1\ 2\ 4)$.

- **Case 2: σ contains at least two 3-cycles.** Write $\sigma = (1\ 2\ 3)(4\ 5\ 6)\mu$. Choose $\tau = (1\ 2\ 4) \in A_n$. Then:

$$\begin{aligned}\sigma_1 &= \tau\sigma\tau^{-1} = (2\ 4\ 3)(1\ 5\ 6)\mu \in N,\\ \sigma' &= \sigma_1\sigma^{-1} = (1\ 2\ 5\ 3\ 4) \in N.\end{aligned}$$

  This produces a 5-cycle in N, which reduces to Case 1 (yielding a 3-cycle in N).

- **Case 3: σ contains one 3-cycle and disjoint 2-cycles.** Write $\sigma = (1\ 2\ 3)(4\ 5)\cdots$. Then $\sigma^2 \in N$. Since the square of any transposition is the identity:

$$\sigma^2 = (1\ 2\ 3)^2 \cdot \mathrm{id} \cdots = (1\ 3\ 2) \in N.$$

  Thus N directly contains the 3-cycle $(1\ 3\ 2)$.

- **Case 4: σ is a product of disjoint 2-cycles.** Write $\sigma = (1\ 2)(3\ 4)\mu$. Let $\tau = (1\ 2\ 3) \in A_n$. Then:

$$\begin{aligned}\sigma_1 &= \tau\sigma\tau^{-1} = (2\ 3)(1\ 4)\mu \in N,\\ \sigma_2 &= \sigma_1\sigma = (2\ 3)(1\ 4)\mu \cdot (1\ 2)(3\ 4)\mu = (1\ 3)(2\ 4) \in N.\end{aligned}$$

  Since $n \geq 5$, choose an element $5 \in \Omega \setminus \{1, 2, 3, 4\}$. Let $\rho = (1\ 3\ 5) \in A_n$. Then:

$$\begin{aligned}\sigma_3 &= \rho\sigma_2\rho^{-1} = (3\ 5)(2\ 4) \in N,\\ \sigma_4 &= \sigma_3\sigma_2 = (3\ 5)(2\ 4)(1\ 3)(2\ 4) = (1\ 3\ 5) \in N.\end{aligned}$$

  Thus N contains the 3-cycle $(1\ 3\ 5)$.

In all possible cases, N contains at least one 3-cycle. By Lemma 3.13, all 3-cycles in A_n are conjugate to this 3-cycle. Since $N \trianglelefteq A_n$, N contains all 3-cycles. By Proposition 3.11(iii), the 3-cycles generate A_n. Therefore, $N = A_n$. Thus A_n has no non-trivial proper normal subgroups, which proves A_n is simple for all $n \geq 5$. □

Theorem 3.15 (Complete Classification of Normal Subgroups of S_n)**.** *The normal subgroups of S_n for all $n \geq 1$ are:*

(1) For $n = 1, 2$: $\{1\}$ and S_n.

(2) For $n = 3$: $\{1\}, A_3, S_3$.

(3) For $n = 4$: $\{1\}, V_4, A_4, S_4$, where $V_4 = \{1, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$.

*(4) For all $n \geq 5$: The **only** normal subgroups of S_n are:*

$$\{1\}, \quad A_n, \quad \text{and} \quad S_n. \tag{3.11}$$

Proof. For $n \geq 5$, let $N \trianglelefteq S_n$. Then $N \cap A_n$ is a normal subgroup of A_n (by Exercise 2 of Unit 2). Since A_n is simple for $n \geq 5$ (Theorem 3.14), there are only two possibilities:

- **Case 1: $N \cap A_n = A_n$.** Then $A_n \subseteq N \subseteq S_n$. By the Correspondence Theorem (Theorem 2.8), subgroups of S_n containing A_n correspond to subgroups of $S_n/A_n \cong \mathbb{Z}_2$. Since $\mathbb{Z}_2$ has only the trivial subgroup and itself, the only possibilities are $N = A_n$ or $N = S_n$.

- **Case 2: $N \cap A_n = \{1\}$.** By the Second Isomorphism Theorem (Theorem 2.6):

$$N/(N \cap A_n) \cong NA_n/A_n \leq S_n/A_n.$$

  Thus $|N| = |NA_n/A_n| \leq [S_n : A_n] = 2$. If $|N| = 1$, then $N = \{1\}$. If $|N| = 2$, then $N = \{1, \sigma\}$ for some odd permutation σ of order 2. Since $N \trianglelefteq S_n$, $\tau\sigma\tau^{-1} = \sigma$ for all $\tau \in S_n$, which means $\sigma \in Z(S_n)$. However, for $n \geq 3$, the center $Z(S_n) = \{1\}$ (since for any non-identity permutation σ, one can choose a transposition not commuting with σ). This is a contradiction. Thus $|N| = 2$ is impossible.

Therefore, for $n \geq 5$, the only normal subgroups of S_n are $\{1\}, A_n$, and S_n. □

3.4 Exercises and Step-by-Step Solutions

Problem 3.16 (Exercise 1: Permutation Cycle Decomposition, Order, and Sign)**.** Write the permutation:

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 3 & 5 & 7 & 8 & 1 & 4 & 2 & 6 \end{pmatrix} \tag{3.12}$$

as a product of disjoint cycles, determine its order $|\sigma|$, and compute its sign $\mathrm{sgn}(\sigma)$.

Solution. 1. **Cycle Decomposition:** Trace the orbits of σ:

- Start with 1: $1 \mapsto 3 \mapsto 7 \mapsto 2 \mapsto 5 \mapsto 1$. This gives the 5-cycle $(1\ 3\ 7\ 2\ 5)$.
- Smallest unvisited element is 4: $4 \mapsto 8 \mapsto 6 \mapsto 4$. This gives the 3-cycle $(4\ 8\ 6)$.

All elements in $\{1, 2, \ldots, 8\}$ have been accounted for. Thus the disjoint cycle decomposition is:

$$\sigma = (1\ 3\ 7\ 2\ 5)(4\ 8\ 6).$$

2. **Order of σ:** By Proposition 3.5:

$$|\sigma| = \mathrm{lcm}(5, 3) = 15.$$

3. **Sign of σ:** Using Theorem 3.9(iv):

$$\begin{aligned} \mathrm{sgn}(\sigma) &= \mathrm{sgn}\big((1\ 3\ 7\ 2\ 5)\big) \cdot \mathrm{sgn}\big((4\ 8\ 6)\big) \\ &= (-1)^{5-1} \cdot (-1)^{3-1} \\ &= (-1)^4 \cdot (-1)^2 \\ &= (+1) \cdot (+1) = +1. \end{aligned}$$

Thus σ is an even permutation ($\sigma \in A_8$).

□

Problem 3.17 (Exercise 2: Conjugacy by Cycle Type in S_n)**.** Prove that two permutations in S_n are conjugate in S_n if and only if they have the same cycle type.

Solution. • **Lemma:** For any cycle $(a_1\ a_2\ \ldots\ a_k)$ and any $\tau \in S_n$:

$$\tau(a_1\ a_2\ \ldots\ a_k)\tau^{-1} = \big(\tau(a_1)\ \tau(a_2)\ \ldots\ \tau(a_k)\big).$$

Proof of Lemma: Let $x \in \{1, \ldots, n\}$. If $x = \tau(a_i)$, then $\tau^{-1}(x) = a_i$, so $(a_1 \ldots a_k)\tau^{-1}(x) = a_{i+1}$, and applying τ gives $\tau(a_{i+1})$. If $x \notin \{\tau(a_1), \ldots, \tau(a_k)\}$, $\tau^{-1}(x) \notin \{a_1, \ldots, a_k\}$, so the cycle fixes $\tau^{-1}(x)$, and $\tau(\tau^{-1}(x)) = x$. Thus the lemma holds.

- **Direction ($\Longrightarrow$):** Suppose $\sigma_1 = \tau\sigma_2\tau^{-1}$. Writing $\sigma_2 = c_1c_2 \cdots c_r$ as a product of disjoint cycles, conjugation distributes across the product:

$$\sigma_1 = (\tau c_1\tau^{-1})(\tau c_2\tau^{-1}) \cdots (\tau c_r\tau^{-1}).$$

By the lemma, each $\tau c_i\tau^{-1}$ is a cycle of the exact same length as c_i. Since τ is a bijection, the resulting cycles have disjoint supports. Thus σ_1 and σ_2 have identical cycle types.

- **Direction ($\Longleftarrow$):** Suppose σ_1 and σ_2 have the same cycle type:

$$\sigma_1 = (a_{11}\ \ldots\ a_{1k_1}) \cdots (a_{r1}\ \ldots\ a_{rk_r}), \quad \sigma_2 = (b_{11}\ \ldots\ b_{1k_1}) \cdots (b_{r1}\ \ldots\ b_{rk_r}).$$

Define the function $\tau : \Omega \to \Omega$ by setting $\tau(a_{ij}) = b_{ij}$ for all i, j, and mapping any fixed points of σ_1 bijectively to the fixed points of σ_2. Then τ is a bijection, so $\tau \in S_n$. By the lemma, $\tau\sigma_1\tau^{-1} = \sigma_2$. Thus σ_1 and σ_2 are conjugate in S_n.

□

Problem 3.18 (Exercise 3: Generation by Adjacent Transpositions)**.** Show that S_n is generated by the set of $n-1$ adjacent transpositions:

$$T = \{(1\ 2),\ (2\ 3),\ (3\ 4),\ \ldots,\ (n-1\ n)\}. \tag{3.13}$$

Solution. By Proposition 3.7, it suffices to show that every transposition $(i\ j)$ with $1 \le i < j \le n$ can be generated by elements of T. We proceed by induction on the distance $d = j - i$:

- *Base Case $d = 1$:* $(i\ i+1) \in T$ by definition.

- *Inductive Step:* Assume all transpositions with distance $< d$ are generated by T. For $j - i = d > 1$, consider the identity:

$$(i\ j) = (j-1\ j)(i\ j-1)(j-1\ j).$$

Evaluating the right side:

$$i \xmapsto{(j-1\ j)} i \xmapsto{(i\ j-1)} j-1 \xmapsto{(j-1\ j)} j,$$
$$j \xmapsto{(j-1\ j)} j-1 \xmapsto{(i\ j-1)} i \xmapsto{(j-1\ j)} i,$$
$$j-1 \xmapsto{(j-1\ j)} j \xmapsto{(i\ j-1)} j \xmapsto{(j-1\ j)} j-1.$$

All other symbols are unchanged. Since $(j-1\ j) \in T$ and $(i\ j-1)$ has distance $d-1 < d$ (which is in $\langle T \rangle$ by the induction hypothesis), it follows that $(i\ j) \in \langle T \rangle$.

Thus every transposition belongs to $\langle T \rangle$, so $\langle T \rangle = S_n$. □

Problem 3.19 (Exercise 4: Normality of the Klein 4-Group in S_4)**.** Prove that the Klein 4-group $V_4 = \{\text{id}, (1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)\}$ is a normal subgroup of S_4.

Solution. 1. V_4 **is a Subgroup:** The identity is in V_4. Each non-identity element has order 2: $((1\ 2)(3\ 4))^2 = \text{id}$. The product of any two distinct non-identity elements yields the third:

$$\big((1\ 2)(3\ 4)\big)\big((1\ 3)(2\ 4)\big) = (1\ 4)(2\ 3) \in V_4.$$

Thus $V_4 \leq S_4$.

2. **Normality in** S_4**:** Notice that $V_4 \setminus \{\text{id}\}$ consists of **all** permutations in S_4 of cycle type $(2,2)$ (products of two disjoint 2-cycles). There are $\frac{1}{2}\binom{4}{2} = 3$ such permutations, and all 3 are in V_4. By Exercise 2, for any $\tau \in S_4$ and any $\sigma \in V_4$, the conjugate $\tau\sigma\tau^{-1}$ must have the same cycle type as σ. Since V_4 contains the entire conjugacy class of $(2,2)$ permutations plus the identity, $\tau V_4 \tau^{-1} = V_4$ for all $\tau \in S_4$. Thus $V_4 \trianglelefteq S_4$.

□

Problem 3.20 (Exercise 5: Non-Existence of Subgroup of Order 6 in A_4)**.** Prove that A_4 has no subgroup of order 6, thereby providing an explicit counterexample to the converse of Lagrange's Theorem.

Solution. The alternating group A_4 has order $|A_4| = 4!/2 = 12$. Its elements are:

- The identity: id (1 element),
- Double transpositions: $(1\ 2)(3\ 4), (1\ 3)(2\ 4), (1\ 4)(2\ 3)$ (3 elements of order 2),
- 3-cycles: $(1\ 2\ 3), (1\ 3\ 2), (1\ 2\ 4), (1\ 4\ 2), (1\ 3\ 4), (1\ 4\ 3), (2\ 3\ 4), (2\ 4\ 3)$ (8 elements of order 3).

Suppose for contradiction that H is a subgroup of A_4 with $|H| = 6$. The index of H in A_4 is:

$$[A_4 : H] = \frac{|A_4|}{|H|} = \frac{12}{6} = 2.$$

By Proposition 2.4, any subgroup of index 2 is normal, so $H \trianglelefteq A_4$.

Since $H \trianglelefteq A_4$, the quotient group A_4/H has order 2. For any 3-cycle $\tau \in A_4$:

$$(\tau H)^2 = \tau^2 H.$$

Since $|A_4/H| = 2$, the square of every element in the quotient group is the identity coset:

$$\tau^2 H = H \implies \tau^2 \in H.$$

Since τ is a 3-cycle, $\tau^3 = \mathrm{id}$, so $\tau = (\tau^2)^2 \in H$. Thus H must contain **every** 3-cycle in A_4. However, there are 8 distinct 3-cycles in A_4. This forces $|H| \geq 8$, which contradicts $|H| = 6$. Therefore, A_4 contains no subgroup of order 6. □

Problem 3.21 (Exercise 6: Transpositions and Subgroups of A_n)**.** Prove that no proper subgroup of S_n containing a transposition can be a subset of A_n.

Solution. By Theorem 3.9(ii), for any transposition $\tau = (i\ j)$, $\mathrm{sgn}(\tau) = -1$. By definition of the alternating group:

$$A_n = \{\sigma \in S_n \mid \mathrm{sgn}(\sigma) = +1\}.$$

Since $\mathrm{sgn}(\tau) = -1 \neq +1$, $\tau \notin A_n$. Therefore, if $H \leq S_n$ and H contains a transposition τ, then $\tau \in H \setminus A_n$, so $H \not\subseteq A_n$. □

Chapter 4

Group Actions, the Class Equation, and Ring Theory Basics

In this chapter, we bridge advanced group theory and commutative ring theory. We develop group actions, orbits, isotropy subgroups, the Orbit-Stabilizer Theorem, Cayley's Theorem, the Class Equation, and the classification of groups of order p^2. We then transition to ring theory: subrings, Gaussian integers, matrix rings, quaternions, ideals, quotient rings, ring homomorphisms, and prime/maximal ideals. Every theorem is presented with a complete, rigorous proof.

4.1 Group Actions, Orbits, and Isotropy Subgroups

Definition 4.1 (Group Action)**.** Let G be a group and let A be a non-empty set. A **group action** (or **action**) of G on A is a map:

$$G \times A \longrightarrow A \tag{4.1}$$

$$(g, a) \longmapsto g \cdot a \tag{4.2}$$

satisfying the following two axioms for all $a \in A$ and $g_1, g_2 \in G$:

(1) Identity: $1_G \cdot a = a$,

(2) Compatibility / Associativity: $g_1 \cdot (g_2 \cdot a) = (g_1 g_2) \cdot a$.

When an action is given, A is called a G**-set**.

Proposition 4.2 (Permutation Representation)**.** *An action of a group G on a set A is equivalent to a group homomorphism:*

$$\sigma : G \longrightarrow S_A, \tag{4.3}$$

where S_A is the symmetric group of all permutations of A, defined by $\sigma(g)(a) = g \cdot a$. The ***kernel*** *of the action is:*

$$\ker(\sigma) = \{g \in G \mid g \cdot a = a \textit{ for all } a \in A\}. \tag{4.4}$$

If $\ker(\sigma) = \{1_G\}$, the action is called ***faithful****.*

Proof. Given an action $(g, a) \mapsto g \cdot a$:

1. For each $g \in G$, define $\sigma_g : A \to A$ by $\sigma_g(a) = g \cdot a$. We check that σ_g is a bijection on A:

   - $(\sigma_g \circ \sigma_{g^{-1}})(a) = g \cdot (g^{-1} \cdot a) = (gg^{-1}) \cdot a = 1_G \cdot a = a$.
   - $(\sigma_{g^{-1}} \circ \sigma_g)(a) = g^{-1} \cdot (g \cdot a) = (g^{-1}g) \cdot a = 1_G \cdot a = a$.

   Thus σ_g is invertible with inverse $\sigma_{g^{-1}}$, so $\sigma_g \in S_A$.

2. Define $\sigma : G \to S_A$ by $\sigma(g) = \sigma_g$. For any $g_1, g_2 \in G$ and $a \in A$:

$$\sigma(g_1g_2)(a) = (g_1g_2) \cdot a = g_1 \cdot (g_2 \cdot a) = \sigma_{g_1}(\sigma_{g_2}(a)) = (\sigma(g_1) \circ \sigma(g_2))(a).$$

Thus $\sigma(g_1g_2) = \sigma(g_1) \circ \sigma(g_2)$, so σ is a homomorphism.

Conversely, given a homomorphism $\sigma : G \to S_A$, defining $g \cdot a = \sigma(g)(a)$ satisfies the action axioms. □

Definition 4.3 (Orbit and Stabilizer / Isotropy Subgroup)**.** Let G act on a set A, and let $a \in A$.

(1) The **orbit** of a, denoted $\mathcal{O}_a$ or $G \cdot a$, is:

$$\mathcal{O}_a = G \cdot a = \{g \cdot a \mid g \in G\} \subseteq A. \tag{4.5}$$

(2) The **stabilizer** (or **isotropy subgroup**) of a, denoted G_a or $\mathrm{Stab}_G(a)$, is:

$$G_a = \mathrm{Stab}_G(a) = \{g \in G \mid g \cdot a = a\} \subseteq G. \tag{4.6}$$

Proposition 4.4 (Orbits Partition the G-Set)**.** *The relation $\sim$ on A defined by $a \sim b \iff \exists g \in G,\ g \cdot a = b$ is an equivalence relation. Consequently, the orbits of G on A form a partition of A:*

$$A = \bigcup_{i \in I} \mathcal{O}_{a_i} \quad (\textit{disjoint union}). \tag{4.7}$$

Proof. We verify the three equivalence relation axioms:

- *Reflexive:* $1_G \cdot a = a \implies a \sim a$.
- *Symmetric:* If $a \sim b$, then $g \cdot a = b$ for some $g \in G$. Acting on both sides by g^{-1}:

  $$g^{-1} \cdot b = g^{-1} \cdot (g \cdot a) = (g^{-1}g) \cdot a = 1_G \cdot a = a.$$

  Thus $b \sim a$.
- *Transitive:* If $a \sim b$ and $b \sim c$, then $g_1 \cdot a = b$ and $g_2 \cdot b = c$. Then:

  $$(g_2g_1) \cdot a = g_2 \cdot (g_1 \cdot a) = g_2 \cdot b = c.$$

  Since $g_2g_1 \in G$, $a \sim c$.

The equivalence classes under $\sim$ are precisely the orbits $\mathcal{O}_a$. Therefore, the orbits partition A. □

Theorem 4.5 (The Orbit-Stabilizer Theorem)**.** *Let G act on a set A, and let $a \in A$. Then $G_a \leq G$, and there is a natural bijection between the orbit $\mathcal{O}_a$ and the set of left cosets G/G_a. In particular, if G is finite:*

$$|\mathcal{O}_a| = [G : G_a] = \frac{|G|}{|G_a|}. \tag{4.8}$$

Proof. 1. **Proof that $G_a \leq G$:**

- *Non-empty:* $1_G \cdot a = a \implies 1_G \in G_a$, so $G_a \neq \emptyset$.

- *Subgroup Criterion:* Let $x, y \in G_a$, so $x \cdot a = a$ and $y \cdot a = a$. From $y \cdot a = a$, act on both sides by y^{-1}:
$$y^{-1} \cdot a = y^{-1} \cdot (y \cdot a) = (y^{-1}y) \cdot a = 1_G \cdot a = a.$$
Now evaluate $(xy^{-1}) \cdot a$:
$$(xy^{-1}) \cdot a = x \cdot (y^{-1} \cdot a) = x \cdot a = a.$$
Thus $xy^{-1} \in G_a$, which proves $G_a \leq G$.

2. **Construction of the Coset Mapping:** Define the map:
$$\begin{aligned} \Phi : G/G_a &\longrightarrow \mathcal{O}_a \\ gG_a &\longmapsto g \cdot a. \end{aligned}$$

3. **Well-Definedness and Injectivity:** For any $g_1, g_2 \in G$:
$$\begin{aligned} g_1 G_a = g_2 G_a &\iff g_2^{-1} g_1 \in G_a \quad \text{(by Lemma 1.18(iii))} \\ &\iff (g_2^{-1} g_1) \cdot a = a \quad \text{(by definition of } G_a) \\ &\iff g_2 \cdot \big((g_2^{-1} g_1) \cdot a\big) = g_2 \cdot a \\ &\iff (g_2 g_2^{-1} g_1) \cdot a = g_2 \cdot a \\ &\iff g_1 \cdot a = g_2 \cdot a \\ &\iff \Phi(g_1 G_a) = \Phi(g_2 G_a). \end{aligned}$$
Reading left-to-right establishes well-definedness; reading right-to-left establishes injectivity.

4. **Surjectivity:** Let $b \in \mathcal{O}_a$. By definition of the orbit, $b = g \cdot a$ for some $g \in G$. Then the coset $gG_a \in G/G_a$ satisfies:
$$\Phi(gG_a) = g \cdot a = b.$$
Thus Φ is surjective.
Since $\Phi : G/G_a \to \mathcal{O}_a$ is a bijection:
$$|\mathcal{O}_a| = |G/G_a| = [G : G_a] = \frac{|G|}{|G_a|}.$$

□

Theorem 4.6 (Cayley's Theorem)**.** *Every group G is isomorphic to a subgroup of the symmetric group S_G. In particular, if $|G| = n$, G is isomorphic to a subgroup of S_n.*

Proof. Let G act on itself ($A = G$) by left multiplication:
$$g \cdot x = gx \quad \text{for all } g, x \in G.$$

1. *Verification of Action Axioms:*
   - $1_G \cdot x = 1_G x = x$.
   - $g_1 \cdot (g_2 \cdot x) = g_1(g_2 x) = (g_1 g_2)x = (g_1 g_2) \cdot x$ (by associativity in G).

2. *Homomorphism:* By Proposition 4.2, this action yields a permutation representation homomorphism:
$$\begin{aligned} \sigma : G &\longrightarrow S_G \\ g &\longmapsto \sigma_g, \quad \text{where } \sigma_g(x) = gx. \end{aligned}$$

3. *Injectivity (Kernel Computation):*

$$\begin{aligned}\ker(\sigma) &= \{g \in G \mid \sigma_g = \mathrm{id}_G\} \\ &= \{g \in G \mid gx = x \text{ for all } x \in G\} \\ &= \{g \in G \mid g \cdot 1_G = 1_G\} \\ &= \{g \in G \mid g = 1_G\} \\ &= \{1_G\}.\end{aligned}$$

Since $\ker(\sigma) = \{1_G\}$, σ is an injective homomorphism. By the First Isomorphism Theorem (Theorem 2.5):

$$G \cong \mathrm{im}(\sigma) \leq S_G.$$

□

4.2 Conjugacy Classes, the Class Equation, and Groups of Order p^2

Definition 4.7 (Conjugation Action and Conjugacy Classes)**.** Let G act on itself by conjugation:

$$g \cdot x = gxg^{-1}. \tag{4.9}$$

(1) The orbits under this action are called the **conjugacy classes** of G. The conjugacy class of $x \in G$ is:

$$\mathcal{C}\ell(x) = \{gxg^{-1} \mid g \in G\}. \tag{4.10}$$

(2) The stabilizer of x is the centralizer $C_G(x)$:

$$G_x = \{g \in G \mid gxg^{-1} = x\} = \{g \in G \mid gx = xg\} = C_G(x). \tag{4.11}$$

(3) By the Orbit-Stabilizer Theorem (Theorem 4.5):

$$|\mathcal{C}\ell(x)| = [G : C_G(x)]. \tag{4.12}$$

Theorem 4.8 (The Class Equation)**.** *Let G be a finite group, and let $g_1, g_2, \ldots, g_r$ be representatives of the distinct conjugacy classes of G that contain strictly more than one element ($g_i \notin Z(G)$). Then:*

$$|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)]. \tag{4.13}$$

Proof. By Proposition 4.4, the conjugacy classes partition the elements of G:

$$G = \bigcup_x \mathcal{C}\ell(x) \implies |G| = \sum_x |\mathcal{C}\ell(x)|.$$

An element x forms a singleton conjugacy class ($|\mathcal{C}\ell(x)| = 1$) if and only if:

$$\mathcal{C}\ell(x) = \{x\} \iff \forall g \in G,\ gxg^{-1} = x \iff \forall g \in G,\ gx = xg \iff x \in Z(G).$$

There are precisely $|Z(G)|$ such elements, and each contributes 1 to the sum of cardinalities. The remaining conjugacy classes have sizes strictly greater than 1. Let $g_1, \ldots, g_r$ be representatives of these distinct classes. By the Orbit-Stabilizer Theorem, each has cardinality:

$$|\mathcal{C}\ell(g_i)| = [G : C_G(g_i)].$$

Summing over all classes gives:

$$|G| = \sum_{z\in Z(G)} 1 + \sum_{i=1}^{r} |\mathcal{C}\ell(g_i)| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)].$$

□

Theorem 4.9 (*p*-Groups Have Non-Trivial Center)**.** *Let p be a prime number. If G is a finite group of order $|G| = p^n$ with $n \geq 1$ (a p-group), then its center is non-trivial:*

$$|Z(G)| \geq p > 1. \tag{4.14}$$

Proof. Apply the Class Equation (Theorem 4.8) to G:

$$|G| = |Z(G)| + \sum_{i=1}^{r} [G : C_G(g_i)].$$

For each representative $g_i \notin Z(G)$, $C_G(g_i)$ is a proper subgroup of G ($C_G(g_i) < G$). By Lagrange's Theorem, $|C_G(g_i)| = p^{m_i}$ with $m_i < n$, which implies the index:

$$[G : C_G(g_i)] = \frac{p^n}{p^{m_i}} = p^{n-m_i}, \quad \text{where } n - m_i \geq 1.$$

Therefore, p divides $[G : C_G(g_i)]$ for all $i \in \{1, 2, \ldots, r\}$. Since $|G| = p^n$ with $n \geq 1$, p also divides $|G|$. Rearranging the Class Equation:

$$|Z(G)| = |G| - \sum_{i=1}^{r} [G : C_G(g_i)].$$

Since p divides every term on the right-hand side, p divides $|Z(G)|$. Since $1_G \in Z(G)$, $|Z(G)| \geq 1$. Because $p \mid |Z(G)|$ and $|Z(G)| \geq 1$, it follows that $|Z(G)| \geq p > 1$. □

Theorem 4.10 (Classification of Groups of Order p^2)**.** *Let p be a prime number. If G is a group of order $|G| = p^2$, then G is abelian. Consequently:*

$$G \cong \mathbb{Z}_{p^2} \quad \textit{or} \quad G \cong \mathbb{Z}_p \times \mathbb{Z}_p. \tag{4.15}$$

Proof. 1. By Theorem 4.9, $|Z(G)| > 1$. By Lagrange's Theorem, $|Z(G)|$ must divide $|G| = p^2$. The positive divisors of p^2 are $1, p, p^2$. Since $|Z(G)| > 1$, we must have:

$$|Z(G)| = p \quad \text{or} \quad |Z(G)| = p^2.$$

2. Suppose for contradiction that $|Z(G)| = p$. Then the order of the quotient group $G/Z(G)$ is:

$$|G/Z(G)| = \frac{|G|}{|Z(G)|} = \frac{p^2}{p} = p.$$

   By Corollary 1.22, any group of prime order is cyclic. Therefore, $G/Z(G)$ is cyclic.

3. By Exercise 3 of Unit 2, if $G/Z(G)$ is cyclic, then G is abelian. If G is abelian, then $Z(G) = G$, so $|Z(G)| = |G| = p^2$. This contradicts our assumption that $|Z(G)| = p$.

4. Therefore, we must have $|Z(G)| = p^2 = |G|$, which implies $G = Z(G)$, so G is abelian.
   Now we classify the abelian group G of order p^2:

- If G contains an element x of order p^2, then $\langle x \rangle = G$, so G is cyclic: $G \cong \mathbb{Z}_{p^2}$.
- If G contains no element of order p^2, then by Lagrange's Theorem every non-identity element has order p. Choose $x \in G \setminus \{1\}$ and $y \in G \setminus \langle x \rangle$. Let $H = \langle x \rangle \cong \mathbb{Z}_p$ and $K = \langle y \rangle \cong \mathbb{Z}_p$. Since G is abelian, $H \trianglelefteq G$ and $K \trianglelefteq G$. Moreover, $H \cap K = \{1\}$ (since $|H \cap K|$ divides p and $y \notin H$). Then $|HK| = \frac{|H||K|}{|H\cap K|} = \frac{p \cdot p}{1} = p^2 = |G|$, so $HK = G$. By Theorem 2.11, $G \cong H \times K \cong \mathbb{Z}_p \times \mathbb{Z}_p$.

□

4.3 Introduction to Rings, Subrings, and Concrete Examples

Definition 4.11 (Ring)**.** A **ring** $(R, +, \cdot)$ is a set R equipped with two binary operations, addition $(+)$ and multiplication $(\cdot)$, satisfying the following axioms for all $a, b, c \in R$:

(1) $(R, +)$ is an abelian group with additive identity denoted 0 (or 0_R), and the additive inverse of a denoted $-a$.

(2) Multiplication is associative: $(a \cdot b) \cdot c = a \cdot (b \cdot c)$.

(3) Multiplication is distributive over addition:

$$a \cdot (b + c) = a \cdot b + a \cdot c \quad \text{and} \quad (a + b) \cdot c = a \cdot c + b \cdot c. \tag{4.16}$$

If $ab = ba$ for all $a, b \in R$, R is called **commutative**. If there exists an element $1 \in R$ $(1 \neq 0)$ such that $1 \cdot a = a \cdot 1 = a$ for all $a \in R$, R is called a **ring with unity** (or ring with identity).

Proposition 4.12 (Elementary Arithmetic in Rings)**.** *Let R be a ring. For all $a, b, c \in R$:*

(i) $0 \cdot a = a \cdot 0 = 0$.

(ii) $(-a)b = a(-b) = -(ab)$.

(iii) $(-a)(-b) = ab$.

(iv) $a(b - c) = ab - ac$ *and* $(a - b)c = ac - bc$.

Proof. **(i)** $0 \cdot a = (0 + 0) \cdot a = 0 \cdot a + 0 \cdot a$. Adding $-(0 \cdot a)$ to both sides: $0 = 0 \cdot a$. Similarly $a \cdot 0 = 0$.

(ii) $ab + (-a)b = (a + (-a))b = 0 \cdot b = 0$. By uniqueness of additive inverses in $(R, +)$, $(-a)b = -(ab)$.

(iii) $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$.

(iv) $a(b - c) = a(b + (-c)) = ab + a(-c) = ab + (-(ac)) = ab - ac$.

□

Definition 4.13 (Subring)**.** Let R be a ring with unity 1_R. A subset $S \subseteq R$ is called a **subring** of R if $1_R \in S$, and for all $x, y \in S$:

$$x - y \in S \quad \text{and} \quad xy \in S. \tag{4.17}$$

4.3.1 Concrete Examples of Rings

Example 4.14 (Gaussian Integers $\mathbb{Z}[i]$)**.** The ring of **Gaussian integers** is:

$$\mathbb{Z}[i] = \{a + bi \mid a, b \in \mathbb{Z},\ i^2 = -1\}. \tag{4.18}$$

It is a commutative ring with unity $1 = 1 + 0i$. The **field norm**:

$$N(a + bi) = a^2 + b^2 = (a + bi)(a - bi) \tag{4.19}$$

satisfies $N(\alpha\beta) = N(\alpha)N(\beta)$. An element $\alpha \in \mathbb{Z}[i]$ is a unit if and only if $N(\alpha) = 1$, giving four units:

$$(\mathbb{Z}[i])^\times = \{1, -1, i, -i\}. \tag{4.20}$$

Example 4.15 (Matrix Rings $M_n(R)$)**.** Let R be a ring. The set $M_n(R)$ of all $n \times n$ matrices with entries in R forms a ring under standard matrix addition and multiplication. For $n \geq 2$, $M_n(R)$ is non-commutative. The identity element is the identity matrix I_n.

Example 4.16 (Real Hamilton Quaternions $\mathbb{H}$)**.** The ring of **quaternions** over $\mathbb{R}$ is:

$$\mathbb{H} = \{a + bi + cj + dk \mid a, b, c, d \in \mathbb{R}\} \tag{4.21}$$

with basis $\{1, i, j, k\}$ and relations:

$$i^2 = j^2 = k^2 = ijk = -1, \quad ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j. \tag{4.22}$$

The conjugate of $q = a + bi + cj + dk$ is $\bar{q} = a - bi - cj - dk$, and the norm is:

$$N(q) = q\bar{q} = a^2 + b^2 + c^2 + d^2. \tag{4.23}$$

Every non-zero quaternion $q \neq 0$ has multiplicative inverse:

$$q^{-1} = \frac{\bar{q}}{N(q)} = \frac{a - bi - cj - dk}{a^2 + b^2 + c^2 + d^2}. \tag{4.24}$$

Thus $\mathbb{H}$ is a non-commutative division ring.

4.4 Ring Homomorphisms, Ideals, and Quotient Rings

Definition 4.17 (Ring Homomorphism)**.** Let R and S be rings with unity. A map $\varphi : R \to S$ is a **ring homomorphism** if for all $a, b \in R$:

- **(i)** $\varphi(a + b) = \varphi(a) + \varphi(b)$,
- **(ii)** $\varphi(ab) = \varphi(a)\varphi(b)$,
- **(iii)** $\varphi(1_R) = 1_S$.

The **kernel** of φ is $\ker(\varphi) = \{r \in R \mid \varphi(r) = 0_S\}$.

Definition 4.18 (Ideal)**.** Let R be a ring. A subset $I \subseteq R$ is called a **(two-sided) ideal**, denoted $I \trianglelefteq R$, if:

(1) $(I, +)$ is a subgroup of $(R, +)$ ($0 \in I$ and $x - y \in I$ for all $x, y \in I$).

(2) Absorption Property: For every $r \in R$ and every $x \in I$:

$$rx \in I \quad \text{and} \quad xr \in I. \tag{4.25}$$

Theorem 4.19 (Quotient Rings and First Isomorphism Theorem for Rings)**.** *Let $I \trianglelefteq R$.*

- ***(i)*** *The set of additive cosets $R/I = \{r + I \mid r \in R\}$ is a ring under the well-defined operations:*

$$(a + I) + (b + I) = (a + b) + I, \tag{4.26}$$
$$(a + I)(b + I) = (ab) + I. \tag{4.27}$$

- ***(ii)*** *If $\varphi : R \to S$ is a ring homomorphism, then $\ker(\varphi) \trianglelefteq R$, and:*

$$R/\ker(\varphi) \cong \operatorname{im}(\varphi). \tag{4.28}$$

Proof. **(i) Well-Definedness of Coset Multiplication:** Suppose $a + I = a' + I$ and $b + I = b' + I$. Then $a' = a + x$ and $b' = b + y$ for some $x, y \in I$. Compute the product $a'b'$:

$$a'b' = (a + x)(b + y) = ab + ay + xb + xy.$$

Since I is an ideal, $y \in I \implies ay \in I$, $x \in I \implies xb \in I$, and $xy \in I$. Since $(I, +)$ is closed under addition, $ay + xb + xy \in I$. Therefore:

$$a'b' \in ab + I \implies a'b' + I = ab + I.$$

Thus multiplication is well-defined. The ring axioms for R/I follow directly from those of R.

(ii) First Isomorphism Theorem for Rings: Let $K = \ker(\varphi)$. Since φ is a group homomorphism on $(R, +)$, K is an additive subgroup. For any $r \in R$ and $k \in K$:

$$\varphi(rk) = \varphi(r)\varphi(k) = \varphi(r) \cdot 0_S = 0_S \implies rk \in K,$$

and similarly $kr \in K$. Thus $K \trianglelefteq R$.

The map $\bar{\varphi} : R/K \to \mathrm{im}(\varphi)$ defined by $\bar{\varphi}(r + K) = \varphi(r)$ is well-defined and bijective on additive groups (by Theorem 2.5). Furthermore:

$$\bar{\varphi}((a + K)(b + K)) = \bar{\varphi}(ab + K) = \varphi(ab) = \varphi(a)\varphi(b) = \bar{\varphi}(a + K)\bar{\varphi}(b + K),$$

and $\bar{\varphi}(1_R + K) = \varphi(1_R) = 1_S$. Thus $\bar{\varphi}$ is a ring isomorphism: $R/\ker(\varphi) \cong \mathrm{im}(\varphi)$.

□

Definition 4.20 (Prime and Maximal Ideals)**.** Let R be a commutative ring with unity $1 \neq 0$, and let $I \subset R$ be a proper ideal ($I \neq R$).

(1) I is a **prime ideal** if for all $a, b \in R$:

$$ab \in I \implies a \in I \quad \text{or} \quad b \in I. \tag{4.29}$$

(2) I is a **maximal ideal** if there is no ideal J of R such that $I \subsetneq J \subsetneq R$.

Theorem 4.21 (Characterization of Prime and Maximal Ideals)**.** *Let R be a commutative ring with unity 1.*

(i) *I is a prime ideal in R if and only if R/I is an integral domain.*

(ii) *I is a maximal ideal in R if and only if R/I is a field.*

(iii) *Every maximal ideal in R is a prime ideal.*

Proof. **(i) Prime Ideals and Integral Domains:** R/I is a commutative ring with unity $1 + I \neq 0 + I$ (since $I \neq R$). The quotient ring R/I is an integral domain if and only if:

$$\begin{aligned}
(a + I)(b + I) = 0 + I &\implies a + I = 0 + I \quad \text{or} \quad b + I = 0 + I \\
&\iff ab + I = I \implies a + I = I \quad \text{or} \quad b + I = I \\
&\iff ab \in I \implies a \in I \quad \text{or} \quad b \in I \\
&\iff I \text{ is a prime ideal in } R.
\end{aligned}$$

(ii) Maximal Ideals and Fields:

- **Direction ($\Longrightarrow$):** Suppose I is maximal. Let $a + I \in R/I$ with $a + I \neq 0 + I$ (so $a \notin I$). Consider the ideal $J = I + (a) = \{i + ra \mid i \in I, r \in R\}$. Since $I \subseteq J$ and $a \in J \setminus I$, we have $I \subsetneq J$. By maximality of I, we must have $J = R$. Since $1 \in R = J$, there exist $i \in I$ and $r \in R$ such that:

$$1 = i + ra \implies 1 - ra = i \in I \implies ra + I = 1 + I \implies (r + I)(a + I) = 1 + I.$$

  Thus $r + I$ is the multiplicative inverse of $a + I$. Hence R/I is a field.

- **Direction ($\Longleftarrow$):** Suppose R/I is a field. Let J be an ideal of R with $I \subsetneq J \subseteq R$. Choose an element $x \in J \setminus I$. Then $x + I \neq 0 + I$ in R/I. Since R/I is a field, $x + I$ is invertible: there exists $y \in R$ such that:

$$(x + I)(y + I) = 1 + I \implies xy + I = 1 + I \implies 1 - xy \in I \subseteq J.$$

  Since $x \in J$ and J is an ideal, $xy \in J$. Then $1 = (1 - xy) + xy \in J$. An ideal containing 1 must be the whole ring: $J = R$. Therefore, I is maximal.

(iii) Maximal $\Longrightarrow$ Prime: If I is maximal, then R/I is a field. Since every field is an integral domain (non-zero elements are units, hence cannot be zero divisors), R/I is an integral domain. By part (i), I is a prime ideal.

□

4.5 Exercises and Step-by-Step Solutions

Problem 4.22 (Exercise 1: Stabilizers in the Same Orbit are Conjugate)**.** Let G act on a set A. Prove that for any $a \in A$ and $g \in G$, the stabilizer of $g \cdot a$ is conjugate to the stabilizer of a:

$$G_{g \cdot a} = gG_a g^{-1}. \tag{4.30}$$

Solution. We prove double containment:

- **Direction ($\subseteq$):** Let $x \in G_{g \cdot a}$. By definition of stabilizer:

$$\begin{aligned} x \cdot (g \cdot a) = g \cdot a &\implies (xg) \cdot a = g \cdot a \\ &\implies g^{-1} \cdot ((xg) \cdot a) = g^{-1} \cdot (g \cdot a) \\ &\implies (g^{-1}xg) \cdot a = (g^{-1}g) \cdot a = 1_G \cdot a = a. \end{aligned}$$

  Thus $g^{-1}xg \in G_a$. Multiplying on the left by g and on the right by g^{-1}:

$$x = g(g^{-1}xg)g^{-1} \in gG_a g^{-1}.$$

  Thus $G_{g \cdot a} \subseteq gG_a g^{-1}$.

- **Direction ($\supseteq$):** Let $y \in gG_a g^{-1}$, so $y = ghg^{-1}$ for some $h \in G_a$ (where $h \cdot a = a$). Evaluate the action of y on $g \cdot a$:

$$\begin{aligned} y \cdot (g \cdot a) &= (ghg^{-1}) \cdot (g \cdot a) \\ &= (ghg^{-1}g) \cdot a \\ &= (gh) \cdot a \\ &= g \cdot (h \cdot a) \\ &= g \cdot a \quad \text{(since } h \in G_a\text{)}. \end{aligned}$$

  Thus $y \in G_{g \cdot a}$, which proves $gG_a g^{-1} \subseteq G_{g \cdot a}$.

Combining both inclusions gives $G_{g\cdot a} = gG_a g^{-1}$. □

Problem 4.23 (Exercise 2: Class Equation of the Dihedral Group D_8)**.** Determine the conjugacy classes and write out the explicit Class Equation for the dihedral group $D_8 = \langle r, s \mid r^4 = s^2 = 1,\ srs = r^{-1}\rangle$ of order 8.

Solution. The elements of D_8 are $\{1, r, r^2, r^3, s, sr, sr^2, sr^3\}$. We compute the conjugates of all elements:

- Identity: $\mathcal{C}\ell(1) = \{1\}$.
- Powers of r:
  - $srs = r^3 \implies \mathcal{C}\ell(r) = \{r, r^3\}$.
  - $sr^2s = (r^2)^{-1} = r^2$, and $rr^2r^{-1} = r^2$. Thus $r^2 \in Z(D_8)$, so $\mathcal{C}\ell(r^2) = \{r^2\}$.

  Thus the center of D_8 is $Z(D_8) = \{1, r^2\}$ (order 2).
- Reflections:
  - $rsr^{-1} = r(sr^{-1}) = r(sr^3) = sr^2 \implies \mathcal{C}\ell(s) = \{s, sr^2\}$.
  - $r(sr)r^{-1} = r(s) = sr^3 \implies \mathcal{C}\ell(sr) = \{sr, sr^3\}$.

The 5 distinct conjugacy classes are:

$$\{1\}, \quad \{r^2\}, \quad \{r, r^3\}, \quad \{s, sr^2\}, \quad \{sr, sr^3\}.$$

The centralizers of the representatives of non-central classes are:

- $C_{D_8}(r) = \langle r\rangle = \{1, r, r^2, r^3\}$ (order 4, index $8/4 = 2$).
- $C_{D_8}(s) = \{1, r^2, s, sr^2\}$ (order 4, index $8/4 = 2$).
- $C_{D_8}(sr) = \{1, r^2, sr, sr^3\}$ (order 4, index $8/4 = 2$).

The explicit Class Equation is:

$$|D_8| = |Z(D_8)| + [D_8 : C_{D_8}(r)] + [D_8 : C_{D_8}(s)] + [D_8 : C_{D_8}(sr)], \tag{4.31}$$

which translates numerically to:

$$8 = 2 + 2 + 2 + 2. \tag{4.32}$$

□

Problem 4.24 (Exercise 3: Alternative Proof that Groups of Order p^2 are Abelian)**.** Prove that any group G of order p^2 (p prime) is abelian without using the Class Equation directly, by analyzing element centralizers.

Solution. By Lagrange's Theorem, the possible orders of elements in G are $1, p, p^2$.

- If G has an element of order p^2, then G is cyclic, hence abelian.
- Now assume every non-identity element of G has order p. Let $x \in G \setminus \{1\}$. The cyclic subgroup $\langle x\rangle$ has order p. The centralizer $C_G(x)$ contains $\langle x\rangle$, so $|C_G(x)| \geq p$. By Lagrange's Theorem, $|C_G(x)|$ divides $|G| = p^2$, so $|C_G(x)| \in \{p, p^2\}$.

  If $|C_G(x)| = p^2$ for all $x \in G$, then $C_G(x) = G$ for all x, meaning every element commutes with all of G, so G is abelian.

  Suppose for contradiction that there exists $x \in G \setminus \{1\}$ such that $|C_G(x)| = p$. Then $C_G(x) = \langle x\rangle$. Since $[G : C_G(x)] = p^2/p = p$, there are p distinct left cosets of $\langle x\rangle$ in G. Since

$\langle x\rangle$ has prime index p in a group of order p^2, $\langle x\rangle \trianglelefteq G$. For any $y \in G \setminus \langle x\rangle$, $yxy^{-1} \in \langle x\rangle$, so $yxy^{-1} = x^k$ for some $1 \le k \le p-1$. Then $y^p x y^{-p} = x^{k^p}$. Since $y^p = 1$, $x = x^{k^p}$, so $k^p \equiv 1 \pmod p$. By Fermat's Little Theorem, $k^p \equiv k \pmod p$, which forces $k \equiv 1 \pmod p$. Thus $yxy^{-1} = x \implies yx = xy$. This means $y \in C_G(x)$, contradicting $y \notin \langle x\rangle = C_G(x)$.

Therefore, G must be abelian. □

Problem 4.25 (Exercise 4: Subring vs. Ideal in Matrix Rings)**.** Prove that the set of upper-triangular matrices:

$$S = \left\{ \begin{pmatrix} a & b \\ 0 & d \end{pmatrix} \middle| \, a, b, d \in \mathbb{R} \right\} \tag{4.33}$$

is a subring of $M_2(\mathbb{R})$, but is **not** an ideal of $M_2(\mathbb{R})$.

Solution. 1. S **is a Subring of** $M_2(\mathbb{R})$**:**

- $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \in S$ (setting $a = 1, b = 0, d = 1$).
- For $A = \begin{pmatrix} a_1 & b_1 \\ 0 & d_1 \end{pmatrix}, B = \begin{pmatrix} a_2 & b_2 \\ 0 & d_2 \end{pmatrix} \in S$:

$$A - B = \begin{pmatrix} a_1 - a_2 & b_1 - b_2 \\ 0 & d_1 - d_2 \end{pmatrix} \in S.$$

- The product is:

$$AB = \begin{pmatrix} a_1 & b_1 \\ 0 & d_1 \end{pmatrix}\begin{pmatrix} a_2 & b_2 \\ 0 & d_2 \end{pmatrix} = \begin{pmatrix} a_1a_2 & a_1b_2 + b_1d_2 \\ 0 & d_1d_2 \end{pmatrix} \in S.$$

Thus S is a subring.

2. S **is Not an Ideal of** $M_2(\mathbb{R})$**:** Choose $X = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \in M_2(\mathbb{R})$ and $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2 \in S$.

Then:

$$XA = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \notin S.$$

Since the bottom-left entry is $1 \neq 0$, $XA \notin S$. Thus S does not satisfy the absorption property, so S is not an ideal.

□

Problem 4.26 (Exercise 5: Prime and Maximal Properties of $(x) \subseteq R[x]$)**.** Let R be a commutative ring with unity 1. Prove that the principal ideal (x) in $R[x]$ is a prime ideal if and only if R is an integral domain, and is a maximal ideal if and only if R is a field.

Solution. Define the evaluation homomorphism at 0:

$$\begin{aligned} \mathrm{ev}_0 : R[x] &\longrightarrow R \\ f(x) = \sum a_i x^i &\longmapsto f(0) = a_0. \end{aligned}$$

- ev_0 is surjective since for any $r \in R$, the constant polynomial $f(x) = r$ satisfies $\mathrm{ev}_0(r) = r$.
- The kernel is:

$$\ker(\mathrm{ev}_0) = \{f(x) \in R[x] \mid f(0) = 0\} = \{a_n x^n + \cdots + a_1 x \mid a_i \in R\} = (x).$$

By the First Isomorphism Theorem for Rings (Theorem 4.19(ii)):

$$R[x]/(x) \cong R.$$

Now apply Theorem 4.21:

- (x) is prime $\iff$ $R[x]/(x)$ is an integral domain $\iff$ R is an integral domain.
- (x) is maximal $\iff$ $R[x]/(x)$ is a field $\iff$ R is a field.

$\square$

Problem 4.27 (Exercise 6: The Ideal (2) in Gaussian Integers)**.** Show that in the ring of Gaussian integers $\mathbb{Z}[i]$, the principal ideal (2) is **not** prime by finding elements $a, b \in \mathbb{Z}[i]$ such that $ab \in (2)$ but $a \notin (2)$ and $b \notin (2)$.

Solution. Consider the element $1 + i \in \mathbb{Z}[i]$. Compute its square:

$$(1 + i)^2 = 1^2 + 2i + i^2 = 1 + 2i - 1 = 2i = 2 \cdot i.$$

Since $i \in \mathbb{Z}[i]$, $2i \in (2)$. Therefore:

$$(1 + i)(1 + i) \in (2).$$

Now we test whether $1 + i \in (2)$. If $1 + i \in (2)$, there would exist $a + bi \in \mathbb{Z}[i]$ such that:

$$1 + i = 2(a + bi) = 2a + 2bi.$$

Equating real parts gives $2a = 1 \implies a = 1/2 \notin \mathbb{Z}$, which is impossible. Thus $1 + i \notin (2)$. Since $(1 + i)(1 + i) \in (2)$ while $1 + i \notin (2)$, the ideal (2) is not prime in $\mathbb{Z}[i]$. $\square$

Chapter 5

Domains, Fields, Polynomial Rings, and Division Algebras

In this concluding chapter, we investigate integral domains, division rings, fields, the rigorous construction of the field of fractions, polynomial rings over fields, the division algorithm, the PID structure of $F[x]$, and the real division algebras $(\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O})$ along with the Cayley-Dickson construction and the classical classification theorems of Frobenius and Hurwitz. Every result is proved with complete mathematical rigor.

5.1 Integral Domains, Division Rings, and Fields

Definition 5.1 (Zero Divisors and Units)**.** Let R be a ring with unity $1 \neq 0$.

(1) A non-zero element $a \in R \setminus \{0\}$ is called a **zero divisor** if there exists a non-zero element $b \in R \setminus \{0\}$ such that:

$$ab = 0 \quad \text{or} \quad ba = 0. \tag{5.1}$$

(2) An element $u \in R$ is called a **unit** (or invertible element) if there exists an element $v \in R$ such that:

$$uv = vu = 1. \tag{5.2}$$

The set of all units forms a group under multiplication, denoted $R^\times$.

Definition 5.2 (Integral Domain, Division Ring, Field)**.** Let R be a ring with unity $1 \neq 0$.

(1) R is an **integral domain** if R is commutative and has no zero divisors:

$$\forall a, b \in R, \quad ab = 0 \implies a = 0 \text{ or } b = 0. \tag{5.3}$$

(2) R is a **division ring** (or **skew field**) if every non-zero element is a unit:

$$R^\times = R \setminus \{0\}. \tag{5.4}$$

(3) R is a **field** if R is a commutative division ring.

Proposition 5.3 (Cancellation Law in Integral Domains)**.** *A commutative ring R with $1 \neq 0$ is an integral domain if and only if the cancellation law holds: for all $a, b, c \in R$ with $a \neq 0$:*

$$ab = ac \implies b = c. \tag{5.5}$$

Proof. • **Direction ($\implies$):** Assume R is an integral domain and $ab = ac$ with $a \neq 0$. Subtracting ac from both sides:

$$ab - ac = 0 \implies a(b - c) = 0.$$

Since R is an integral domain, it has no zero divisors. Since $a \neq 0$, we must have $b - c = 0$, which implies $b = c$.

- **Direction ($\Longleftarrow$):** Assume the cancellation law holds in R. Suppose $ab = 0$ with $a \neq 0$. Using Proposition 4.12(i), we can write:

$$ab = a \cdot 0.$$

Since $a \neq 0$, applying the cancellation law gives $b = 0$. Thus R contains no zero divisors, so R is an integral domain.

□

Theorem 5.4 (Finite Integral Domains are Fields)**.** *Every finite integral domain is a field.*

Proof. Let D be a finite integral domain and let $a \in D$ with $a \neq 0$. We must show that a has a multiplicative inverse in D.

Define the left-multiplication map:

$$\begin{aligned} \lambda_a : D &\longrightarrow D \\ x &\longmapsto ax. \end{aligned}$$

1. **Injectivity:** Suppose $\lambda_a(x) = \lambda_a(y)$ for some $x, y \in D$. Then:

$$ax = ay \implies ax - ay = 0 \implies a(x - y) = 0.$$

Since D is an integral domain and $a \neq 0$, we must have $x - y = 0 \implies x = y$. Thus λ_a is injective.

2. **Surjectivity by Pigeonhole Principle:** Since D is a finite set, any injective map from D to itself is automatically surjective. Thus $\lambda_a(D) = D$.

3. **Existence of Multiplicative Inverse:** Since $1 \in D$, the identity element must lie in the image of λ_a. Therefore, there exists some element $b \in D$ such that:

$$\lambda_a(b) = ab = 1.$$

Since D is commutative, $ba = ab = 1$, which proves that $b = a^{-1} \in D$.

Since every non-zero element $a \in D$ is invertible, D is a field. □

5.2 The Field of Fractions

Let D be an integral domain. We construct the smallest field containing an isomorphic copy of D.

Definition 5.5 (Construction of the Field of Fractions)**.** Let $S = D \times (D \setminus \{0\})$. Define a relation $\sim$ on S by:

$$(a, b) \sim (c, d) \iff ad = bc. \tag{5.6}$$

Lemma 5.6 (Equivalence Relation for Fractions)**.** *The relation $\sim$ is an equivalence relation on S.*

Proof.
- *Reflexive:* $ab = ba \implies (a, b) \sim (a, b)$.
- *Symmetric:* $(a, b) \sim (c, d) \implies ad = bc \implies cb = da \implies (c, d) \sim (a, b)$.

- *Transitive:* Suppose $(a, b) \sim (c, d)$ and $(c, d) \sim (e, f)$. Then $ad = bc$ and $cf = de$. Multiply the first equation by f and the second by b:

$$adf = bcf \quad \text{and} \quad bcf = bde.$$

Equating the two expressions gives:

$$adf = bde \implies (af - be)d = 0.$$

Since $(c, d) \in S$, $d \neq 0$. Because D is an integral domain and $d \neq 0$, we must have:

$$af - be = 0 \implies af = be \implies (a, b) \sim (e, f).$$

Thus $\sim$ is an equivalence relation. □

Definition 5.7 (Field of Fractions)**.** The set of equivalence classes of S under $\sim$ is denoted:

$$Q(D) = \mathrm{Frac}(D) = \left\{ \frac{a}{b} \;\middle|\; a \in D,\, b \in D \setminus \{0\} \right\}, \tag{5.7}$$

where $\frac{a}{b}$ denotes the equivalence class $[(a, b)]_\sim$.

Theorem 5.8 (Field Structure on $Q(D)$)**.** *$Q(D)$ is a field under the well-defined operations:*

$$\frac{a}{b} + \frac{c}{d} = \frac{ad + bc}{bd}, \tag{5.8}$$

$$\frac{a}{b} \cdot \frac{c}{d} = \frac{ac}{bd}. \tag{5.9}$$

The additive identity is $\frac{0}{1}$, the multiplicative identity is $\frac{1}{1}$, and the inverse of $\frac{a}{b} \neq \frac{0}{1}$ is $\left(\frac{a}{b}\right)^{-1} = \frac{b}{a}$. Moreover, the mapping:

$$\iota : D \hookrightarrow Q(D) \tag{5.10}$$

$$a \longmapsto \frac{a}{1} \tag{5.11}$$

is an injective ring homomorphism.

Proof. 1. **Well-Definedness of Addition:** Suppose $(a, b) \sim (a', b')$ and $(c, d) \sim (c', d')$, so $ab' = a'b$ and $cd' = c'd$. We must show:

$$\begin{aligned}(ad + bc, bd) \sim (a'd' + b'c', b'd') &\iff (ad + bc)b'd' = (a'd' + b'c')bd \\ &\iff adb'd' + bcb'd' = a'd'bd + b'c'bd \\ &\iff (ab')dd' + (cd')bb' = (a'b)dd' + (c'd)bb'.\end{aligned}$$

Since $ab' = a'b$ and $cd' = c'd$, the two sides are identical. Thus addition is well-defined.

2. **Well-Definedness of Multiplication:** Similarly, $(ac)b'd' = (ab')(cd') = (a'b)(c'd) = (a'c')bd$, so multiplication is well-defined.

3. **Field Axioms:** Associativity, commutativity, and distributivity follow directly by polynomial algebra in D. The additive inverse of $\frac{a}{b}$ is $\frac{-a}{b}$. For any non-zero fraction $\frac{a}{b} \neq \frac{0}{1}$, $a \neq 0$. Then $\frac{b}{a} \in Q(D)$, and $\frac{a}{b} \cdot \frac{b}{a} = \frac{ab}{ba} = \frac{1}{1}$. Thus $Q(D)$ is a field.

4. **Canonical Embedding ι:** $\iota(a + b) = \frac{a+b}{1} = \frac{a}{1} + \frac{b}{1} = \iota(a) + \iota(b)$, and $\iota(ab) = \frac{ab}{1} = \iota(a)\iota(b)$, with $\iota(1) = \frac{1}{1}$. $\ker(\iota) = \{a \in D \mid \frac{a}{1} = \frac{0}{1}\} = \{a \in D \mid a \cdot 1 = 0 \cdot 1\} = \{0\}$. Thus ι is injective. □

Theorem 5.9 (Universal Mapping Property of the Field of Fractions)**.** *Let D be an integral domain, and let K be any field. If $\varphi : D \hookrightarrow K$ is an injective ring homomorphism, there exists a **unique** field homomorphism $\Phi : Q(D) \to K$ such that $\Phi \circ \iota = \varphi$, defined by:*

$$\Phi\left(\frac{a}{b}\right) = \varphi(a)\big(\varphi(b)\big)^{-1}. \tag{5.12}$$

Proof. 1. **Well-Definedness:** If $\frac{a}{b} = \frac{c}{d}$, then $ad = bc$. Applying the homomorphism φ:

$$\varphi(ad) = \varphi(bc) \implies \varphi(a)\varphi(d) = \varphi(b)\varphi(c).$$

Since φ is injective and $b, d \neq 0$, $\varphi(b), \varphi(d) \neq 0$ in the field K, and thus are invertible. Multiplying by $\big(\varphi(b)\big)^{-1}\big(\varphi(d)\big)^{-1}$:

$$\varphi(a)\big(\varphi(b)\big)^{-1} = \varphi(c)\big(\varphi(d)\big)^{-1}.$$

Thus Φ is well-defined.

2. **Homomorphism Property:**

$$\begin{aligned}\Phi\left(\frac{a}{b} + \frac{c}{d}\right) &= \Phi\left(\frac{ad+bc}{bd}\right) = \varphi(ad+bc)\big(\varphi(bd)\big)^{-1} \\ &= \big(\varphi(a)\varphi(d) + \varphi(b)\varphi(c)\big)\big(\varphi(b)\varphi(d)\big)^{-1} \\ &= \varphi(a)\big(\varphi(b)\big)^{-1} + \varphi(c)\big(\varphi(d)\big)^{-1} \\ &= \Phi\left(\frac{a}{b}\right) + \Phi\left(\frac{c}{d}\right).\end{aligned}$$

Similarly, $\Phi\left(\frac{a}{b} \cdot \frac{c}{d}\right) = \Phi\left(\frac{a}{b}\right)\Phi\left(\frac{c}{d}\right)$ and $\Phi\left(\frac{1}{1}\right) = 1_K$.

3. **Commutativity of Diagram:** For any $a \in D$, $\Phi(\iota(a)) = \Phi\left(\frac{a}{1}\right) = \varphi(a)\big(\varphi(1)\big)^{-1} = \varphi(a) \cdot 1_K^{-1} = \varphi(a)$.

4. **Uniqueness:** Any field homomorphism $\Psi : Q(D) \to K$ satisfying $\Psi \circ \iota = \varphi$ must satisfy $\Psi\left(\frac{a}{1}\right) = \varphi(a)$ and $\Psi\left(\frac{b}{1}\right) = \varphi(b)$. Since $\frac{a}{b} = \frac{a}{1} \cdot \left(\frac{b}{1}\right)^{-1}$:

$$\Psi\left(\frac{a}{b}\right) = \Psi\left(\frac{a}{1}\right)\left(\Psi\left(\frac{b}{1}\right)\right)^{-1} = \varphi(a)\big(\varphi(b)\big)^{-1} = \Phi\left(\frac{a}{b}\right).$$

□

5.3 Polynomial Rings, Division Algorithm, and Ideals over a Field

Definition 5.10 (Polynomial Ring)**.** Let R be a commutative ring with unity 1. The **polynomial ring** in the indeterminate x with coefficients in R, denoted $R[x]$, consists of all formal expressions:

$$f(x) = \sum_{i=0}^{n} a_i x^i = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0, \quad a_i \in R. \tag{5.13}$$

If $a_n \neq 0$, the **degree** of $f(x)$ is $\deg(f) = n$, and a_n is the **leading coefficient**. If $a_n = 1$, $f(x)$ is called **monic**. The zero polynomial $f(x) = 0$ has degree $\deg(0) = -\infty$.

Proposition 5.11 (Degree Properties in Integral Domains)**.** *If R is an integral domain and $f(x), g(x) \in R[x] \setminus \{0\}$, then:*

$$\deg(fg) = \deg(f) + \deg(g). \tag{5.14}$$

Consequently, $R[x]$ is an integral domain, and $(R[x])^\times = R^\times$.

Proof. Let $f(x) = a_n x^n + \cdots + a_0$ with $a_n \neq 0$ ($\deg(f) = n$), and $g(x) = b_m x^m + \cdots + b_0$ with $b_m \neq 0$ ($\deg(g) = m$). In the product $f(x)g(x)$, the coefficient of the highest power x^{n+m} is $a_n b_m$. Since R is an integral domain and $a_n \neq 0, b_m \neq 0$, we have $a_n b_m \neq 0$. Therefore, $\deg(fg) = n + m = \deg(f) + \deg(g)$.

If $fg = 0$, then $f = 0$ or $g = 0$ (otherwise $\deg(fg) \geq 0$, a contradiction). Thus $R[x]$ is an integral domain. If $f(x) \in (R[x])^\times$, there exists $g(x)$ such that $fg = 1$. Then $\deg(f) + \deg(g) = \deg(1) = 0 \implies \deg(f) = 0$, so $f(x) \in R^\times$. □

Theorem 5.12 (Division Algorithm for Polynomials over a Field)**.** *Let F be a field, and let $f(x), g(x) \in F[x]$ with $g(x) \neq 0$. Then there exist* ***unique*** *polynomials $q(x), r(x) \in F[x]$ such that:*

$$f(x) = q(x)g(x) + r(x), \quad \textit{where } r(x) = 0 \textit{ or } \deg(r) < \deg(g). \tag{5.15}$$

Proof.

- **Existence (Strong Induction on $\deg(f)$):** Let $m = \deg(g) \geq 0$. If $f(x) = 0$ or $\deg(f) < m$, we set $q(x) = 0$ and $r(x) = f(x)$.

  Now assume $n = \deg(f) \geq m$, and assume the result holds for all polynomials of degree $< n$. Write $f(x) = a_n x^n + \cdots + a_0$ and $g(x) = b_m x^m + \cdots + b_0$ with $a_n, b_m \neq 0$. Since F is a field, $b_m^{-1} \in F$. Define:

  $$f_1(x) = f(x) - \left(\frac{a_n}{b_m} x^{n-m}\right) g(x).$$

  The coefficient of x^n in $f_1(x)$ is $a_n - \left(\frac{a_n}{b_m}\right) b_m = a_n - a_n = 0$. Thus either $f_1(x) = 0$ or $\deg(f_1) < n$. By the induction hypothesis, there exist $q_1(x), r(x) \in F[x]$ such that:

  $$f_1(x) = q_1(x)g(x) + r(x), \quad \text{with } r(x) = 0 \text{ or } \deg(r) < m.$$

  Substituting back into $f(x)$:

  $$\begin{aligned} f(x) &= \left(\frac{a_n}{b_m} x^{n-m}\right) g(x) + f_1(x) \\ &= \left(\frac{a_n}{b_m} x^{n-m}\right) g(x) + q_1(x)g(x) + r(x) \\ &= \left(\frac{a_n}{b_m} x^{n-m} + q_1(x)\right) g(x) + r(x). \end{aligned}$$

  Setting $q(x) = \frac{a_n}{b_m} x^{n-m} + q_1(x)$ completes the existence proof.

- **Uniqueness:** Suppose $f(x) = q_1(x)g(x) + r_1(x) = q_2(x)g(x) + r_2(x)$. Rearranging:

  $$(q_1(x) - q_2(x))g(x) = r_2(x) - r_1(x).$$

  If $q_1(x) - q_2(x) \neq 0$, then:

  $$\deg\Big((q_1 - q_2)g\Big) = \deg(q_1 - q_2) + \deg(g) \geq \deg(g).$$

  However, the right side satisfies:

  $$\deg(r_2 - r_1) \leq \max(\deg(r_1), \deg(r_2)) < \deg(g),$$

  a contradiction. Therefore, $q_1(x) - q_2(x) = 0 \implies q_1(x) = q_2(x)$, which forces $r_1(x) = r_2(x)$. □

Theorem 5.13 ($F[x]$ is a Principal Ideal Domain (P.I.D.))**.** *Let F be a field. Every ideal $I \subseteq F[x]$ is principal. That is, there exists a unique monic polynomial $p(x) \in F[x]$ (or $p(x) = 0$ if $I = \{0\}$) such that:*

$$I = (p(x)) = \{q(x)p(x) \mid q(x) \in F[x]\}. \tag{5.16}$$

Proof. 1. If $I = \{0\}$, then $I = (0)$, which is principal.

2. If $I \neq \{0\}$, the set of degrees $S = \{\deg(f) \mid f(x) \in I,\ f(x) \neq 0\}$ is a non-empty subset of non-negative integers $\mathbb{N} \cup \{0\}$. By the Well-Ordering Principle, S has a minimum element $d \geq 0$. Choose a non-zero polynomial $g(x) \in I$ with $\deg(g) = d$. Let $c \in F^\times$ be the leading coefficient of $g(x)$. The polynomial $p(x) = c^{-1}g(x)$ is monic, belongs to I (since I is an ideal), and has degree d.

3. Since $p(x) \in I$, the principal ideal $(p(x)) \subseteq I$.

4. Now let $f(x) \in I$ be arbitrary. By the Division Algorithm (Theorem 5.12):

$$f(x) = q(x)p(x) + r(x), \quad \text{where } r(x) = 0 \text{ or } \deg(r) < \deg(p) = d.$$

   Rearranging gives $r(x) = f(x) - q(x)p(x)$. Since $f(x) \in I$ and $p(x) \in I$, we have $r(x) \in I$. If $r(x) \neq 0$, then $\deg(r) < d$, which contradicts the minimality of $d = \deg(p)$. Therefore, $r(x) = 0$.

5. Thus $f(x) = q(x)p(x) \in (p(x))$, which proves $I \subseteq (p(x))$.
   Combining both inclusions gives $I = (p(x))$. □

Theorem 5.14 (Maximal Ideals and Irreducibility in $F[x]$)**.** *Let F be a field and let $p(x) \in F[x]$ be a non-constant polynomial ($\deg(p) \geq 1$). The following statements are equivalent:*

(1) *$p(x)$ is irreducible over F.*

(2) *$(p(x))$ is a maximal ideal in $F[x]$.*

(3) *$(p(x))$ is a prime ideal in $F[x]$.*

(4) *The quotient ring $F[x]/(p(x))$ is a field.*

Proof. • (2) $\iff$ (4) and (2) $\implies$ (3) hold for any commutative ring by Theorem 4.21.

- **Proof that** (1) $\implies$ (2)**:** Assume $p(x)$ is irreducible over F. Let J be an ideal of $F[x]$ such that $(p(x)) \subseteq J \subseteq F[x]$. Since $F[x]$ is a PID (Theorem 5.13), $J = (g(x))$ for some $g(x) \in F[x]$. Since $p(x) \in J = (g(x))$, $g(x)$ divides $p(x)$, so:

$$p(x) = g(x)h(x) \quad \text{for some } h(x) \in F[x].$$

  Since $p(x)$ is irreducible, one of the factors must be a unit in $F[x]$ (a non-zero constant in $F^\times$):

  - If $g(x) \in F^\times$, then $J = (g(x)) = F[x]$.
  - If $h(x) \in F^\times$, then $g(x) = h(x)^{-1}p(x) \in (p(x))$, so $J = (g(x)) = (p(x))$.

  Thus there are no proper ideals strictly containing $(p(x))$, which proves $(p(x))$ is a maximal ideal.

- **Proof that** (3) $\implies$ (1)**:** Assume $(p(x))$ is a prime ideal. Suppose $p(x) = a(x)b(x)$ for some $a(x), b(x) \in F[x]$. Then $a(x)b(x) \in (p(x))$. Since $(p(x))$ is a prime ideal:

$$a(x) \in (p(x)) \quad \text{or} \quad b(x) \in (p(x)).$$

If $a(x) \in (p(x))$, then $a(x) = k(x)p(x)$ for some $k(x) \neq 0$. Then $\deg(a) \geq \deg(p)$. Since $\deg(p) = \deg(a) + \deg(b)$, this forces:

$$\deg(b) = 0 \implies b(x) \in F^\times \text{ is a unit.}$$

Similarly, if $b(x) \in (p(x))$, $a(x)$ is a unit. Therefore, $p(x)$ is irreducible over F.

□

5.4 The Real Division Algebras: Reals, Complex, Quaternions, and Octonions

Definition 5.15 (Algebra over a Field)**.** An **algebra** A over a field F is a vector space A over F equipped with a bilinear multiplication $A \times A \to A$. An algebra A is a **division algebra** if for every $a, b \in A$ with $a \neq 0$, the equations $ax = b$ and $ya = b$ have unique solutions $x, y \in A$.

5.4.1 The Cayley-Dickson Construction

The classical sequence of division algebras over $\mathbb{R}$ is obtained via iterative doubling (the **Cayley-Dickson process**):

$$\mathbb{R} \xrightarrow{\text{dim } 1} \mathbb{C} \xrightarrow{\text{dim } 2} \mathbb{H} \xrightarrow{\text{dim } 4} \mathbb{O} \xrightarrow{\text{dim } 8} \tag{5.17}$$

At each stage, an algebra with involution $(A, \bar{\cdot})$ is doubled to $A' = A \oplus A\ell$:

$$(a, b)(c, d) = (ac - \bar{d}b,\ da + b\bar{c}), \tag{5.18}$$

$$\overline{(a, b)} = (\bar{a},\ -b), \tag{5.19}$$

$$N(a, b) = (a, b)\overline{(a, b)} = N(a) + N(b). \tag{5.20}$$

Algebra	Dim / $\mathbb{R}$	Commutative?	Associative?	Division Algebra?
$\mathbb{R}$ (Real Numbers)	1	Yes	Yes	Yes
$\mathbb{C}$ (Complex Numbers)	2	Yes	Yes	Yes
$\mathbb{H}$ (Quaternions)	4	**No**	Yes	Yes
$\mathbb{O}$ (Octonions)	8	**No**	**No** (Alternative)	Yes

Table 5.1: The Cayley-Dickson Hierarchy of Real Division Algebras.

5.4.2 Properties of the Four Algebras

(1) Real Numbers $\mathbb{R}$: 1-dimensional, commutative, associative, ordered field.

(2) Complex Numbers $\mathbb{C} = \mathbb{R} \oplus \mathbb{R}i$: 2-dimensional algebra with $i^2 = -1$. Commutative and associative. Conjugate: $\overline{a + bi} = a - bi$. Multiplicative norm: $|z|^2 = z\bar{z} = a^2 + b^2$.

(3) Quaternions $\mathbb{H} = \mathbb{C} \oplus \mathbb{C}j$: 4-dimensional algebra with basis $\{1, i, j, k\}$:

$$i^2 = j^2 = k^2 = ijk = -1, \quad ij = -ji = k, \quad jk = -kj = i, \quad ki = -ik = j. \tag{5.21}$$

Conjugate: $\bar{q} = a - bi - cj - dk$. Multiplicative norm: $N(q) = q\bar{q} = a^2 + b^2 + c^2 + d^2$. Every non-zero quaternion is invertible with $q^{-1} = \frac{\bar{q}}{N(q)}$. Non-commutative, but associative.

(4) Octonions $\mathbb{O} = \mathbb{H} \oplus \mathbb{H}\ell$: 8-dimensional algebra with basis $\{e_0 = 1, e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$. Multiplication is governed by the **Fano Plane** projective geometry. $\mathbb{O}$ is **non-associative**: $(xy)z \neq x(yz)$ in general. However, it satisfies the **alternative property**:

$$x(xy) = (xx)y \quad \text{and} \quad (yx)x = y(xx) \quad \text{for all } x, y \in \mathbb{O}. \tag{5.22}$$

The norm satisfies $N(xy) = N(x)N(y)$, which implies $xy = 0 \iff x = 0$ or $y = 0$, making $\mathbb{O}$ a normed alternative division algebra.

5.4.3 The Fundamental Classification Theorems

Theorem 5.16 (Frobenius' Theorem (1877))**.** *Every finite-dimensional associative division algebra over $\mathbb{R}$ is isomorphic to one of:*

$$\mathbb{R}, \quad \mathbb{C}, \quad \textit{or} \quad \mathbb{H}. \tag{5.23}$$

Theorem 5.17 (Hurwitz's Theorem (1898))**.** *Every finite-dimensional normed division algebra over $\mathbb{R}$ is isomorphic to one of:*

$$\mathbb{R}, \quad \mathbb{C}, \quad \mathbb{H}, \quad \textit{or} \quad \mathbb{O}. \tag{5.24}$$

5.5 Exercises and Step-by-Step Solutions

Problem 5.18 (Exercise 1: The Quotient Ring $\mathbb{R}[x]/(x^2+1)$)**.** Prove that the polynomial $x^2 + 1$ is irreducible over $\mathbb{R}$, and prove that the quotient ring $\mathbb{R}[x]/(x^2+1)$ is isomorphic to the field of complex numbers $\mathbb{C}$.

Solution. 1. **Irreducibility of $x^2 + 1$ over $\mathbb{R}$:** Since $\deg(x^2+1) = 2$, $x^2 + 1$ is reducible over $\mathbb{R}$ if and only if it has a root in $\mathbb{R}$. For any $r \in \mathbb{R}$, $r^2 \geq 0$, so $r^2 + 1 \geq 1 > 0$. Thus $x^2 + 1$ has no real roots, so it is irreducible over $\mathbb{R}$.

2. **Isomorphism with $\mathbb{C}$:** Define the evaluation homomorphism at $i \in \mathbb{C}$:

$$\begin{aligned} \varphi : \mathbb{R}[x] &\longrightarrow \mathbb{C} \\ f(x) &\longmapsto f(i). \end{aligned}$$

- *Surjectivity:* For any $a + bi \in \mathbb{C}$, the polynomial $g(x) = a + bx \in \mathbb{R}[x]$ satisfies $\varphi(g(x)) = a + bi$. Thus φ is surjective.
- *Kernel Calculation:* $\varphi(x^2+1) = i^2 + 1 = -1 + 1 = 0 \implies (x^2+1) \subseteq \ker(\varphi)$. Since $x^2 + 1$ is irreducible and $\mathbb{R}[x]$ is a PID, the ideal (x^2+1) is maximal (Theorem 5.14). Since $\ker(\varphi)$ is a proper ideal ($\varphi(1) = 1 \neq 0$), maximality forces $\ker(\varphi) = (x^2+1)$.

By the First Isomorphism Theorem for Rings (Theorem 4.19(ii)):

$$\mathbb{R}[x]/(x^2+1) \cong \mathbb{C}.$$

□

Problem 5.19 (Exercise 2: Non-Zero Prime Ideals are Maximal in a PID)**.** Prove that in any Principal Ideal Domain (PID), every non-zero prime ideal is a maximal ideal.

Solution. Let R be a PID and let $P = (p)$ be a non-zero prime ideal ($p \neq 0$). Let $M = (m)$ be an ideal of R such that $(p) \subseteq (m) \subseteq R$. We must show that $(m) = (p)$ or $(m) = R$.

- Since $p \in (m)$, m divides p, so there exists $r \in R$ such that $p = mr$.

- Since (p) is a prime ideal and $mr = p \in (p)$, we have $m \in (p)$ or $r \in (p)$.
  - If $m \in (p)$, then $(m) \subseteq (p)$, so $(m) = (p)$.
  - If $r \in (p)$, then $r = kp$ for some $k \in R$. Then $p = mr = m(kp) = (mk)p \implies (1 - mk)p = 0$. Since R is an integral domain and $p \neq 0$, $1 - mk = 0 \implies mk = 1$. Thus $m \in R^\times$ is a unit, so $(m) = R$.

Thus there are no proper ideals strictly containing (p), which proves (p) is maximal. □

Problem 5.20 (Exercise 3: Euclidean Algorithm and Bézout Identity in $\mathbb{Q}[x]$)**.** Use the Euclidean Algorithm in $\mathbb{Q}[x]$ to compute the greatest common divisor of:

$$f(x) = x^4 - 2x^3 - 3x^2 + 4x + 4 \quad \text{and} \quad g(x) = x^3 - x^2 - 4x + 4, \tag{5.25}$$

and express $\gcd(f, g)$ as a linear combination $a(x)f(x) + b(x)g(x)$.

Solution. 1. **Step 1: Divide $f(x)$ by $g(x)$ via Polynomial Long Division:** Dividing $f(x) = x^4 - 2x^3 - 3x^2 + 4x + 4$ by $g(x) = x^3 - x^2 - 4x + 4$:

$$\begin{aligned} f(x) - xg(x) &= (x^4 - 2x^3 - 3x^2 + 4x + 4) - (x^4 - x^3 - 4x^2 + 4x) \\ &= -x^3 + x^2 + 4. \end{aligned}$$

Next, $(-x^3 + x^2 + 4) - (-1)g(x) = (-x^3 + x^2 + 4) - (-x^3 + x^2 + 4x - 4) = -4x + 8$. Thus the quotient is $q_1(x) = x - 1$ and the remainder is $r_1(x) = -4x + 8 = -4(x - 2)$:

$$f(x) = (x - 1)g(x) + (-4x + 8).$$

2. **Step 2: Divide $g(x)$ by the monic remainder $x - 2$:** Dividing $g(x) = x^3 - x^2 - 4x + 4$ by $x - 2$:

$$\begin{aligned} g(x) &= x^2(x - 1) - 4(x - 1) = (x^2 - 4)(x - 1) \\ &= (x - 2)(x + 2)(x - 1) \\ &= (x^2 + x - 2)(x - 2) + 0. \end{aligned}$$

Since the remainder is 0, the last non-zero monic remainder is:

$$\gcd\big(f(x), g(x)\big) = x - 2.$$

3. **Step 3: Determine the Bézout Identity:** From Step 1, we have:

$$-4(x - 2) = f(x) - (x - 1)g(x).$$

Dividing by -4:

$$x - 2 = -\frac{1}{4}f(x) + \left(\frac{x - 1}{4}\right)g(x).$$

Thus $a(x) = -\frac{1}{4}$ and $b(x) = \frac{1}{4}x - \frac{1}{4}$.

□

Problem 5.21 (Exercise 4: Polynomial Rings over Domains are Never Fields)**.** Prove that if D is an integral domain, then the polynomial ring $D[x]$ is never a field.

Solution. Consider the indeterminate element $x \in D[x]$. Since $\deg(x) = 1 \geq 1$, $x \neq 0$. Suppose for contradiction that $D[x]$ is a field. Then every non-zero element must have a multiplicative inverse, so there exists $g(x) \in D[x]$ such that:

$$x \cdot g(x) = 1.$$

Taking degrees of both sides using Proposition 5.11:

$$\deg(x) + \deg(g) = \deg(1) \implies 1 + \deg(g) = 0 \implies \deg(g) = -1.$$

However, the degree of any non-zero polynomial in $D[x]$ must be a non-negative integer ≥ 0. This is a contradiction. Therefore, x is not invertible in $D[x]$, so $D[x]$ cannot be a field. $\square$

Problem 5.22 (Exercise 5: Computation in Hamilton Quaternions $\mathbb{H}$)**.** Let $q = 1+2i-j+3k \in \mathbb{H}$. Compute the norm $N(q)$, the conjugate $\bar{q}$, and the multiplicative inverse q^{-1}.

Solution. Given $q = a + bi + cj + dk$ with $a = 1, b = 2, c = -1, d = 3$:

1. **Conjugate $\bar{q}$:**
$$\bar{q} = a - bi - cj - dk = 1 - 2i + j - 3k.$$

2. **Norm $N(q)$:**
$$N(q) = a^2 + b^2 + c^2 + d^2 = 1^2 + 2^2 + (-1)^2 + 3^2 = 1 + 4 + 1 + 9 = 15.$$

3. **Inverse q^{-1}:**
$$q^{-1} = \frac{\bar{q}}{N(q)} = \frac{1 - 2i + j - 3k}{15} = \frac{1}{15} - \frac{2}{15}i + \frac{1}{15}j - \frac{1}{5}k.$$
Verification: $qq^{-1} = \frac{(1+2i-j+3k)(1-2i+j-3k)}{15} = \frac{15}{15} = 1.$

$\square$

Problem 5.23 (Exercise 6: Non-Associativity of the Octonions $\mathbb{O}$)**.** Using the Cayley-Dickson multiplication table, verify that the octonions $\mathbb{O}$ are non-associative by calculating $(e_1e_2)e_4$ and $e_1(e_2e_4)$.

Solution. Using the standard Fano plane orientation for imaginary basis units $e_1, \ldots, e_7$ of $\mathbb{O}$:

- From the Fano cycle (e_1, e_2, e_3), we have $e_1e_2 = e_3$. Then:
$$(e_1e_2)e_4 = e_3e_4.$$
From the line (e_3, e_4, e_7), $e_3e_4 = e_7$. Thus $(e_1e_2)e_4 = e_7$.

- On the other hand, from the line (e_2, e_4, e_6), $e_2e_4 = e_6$. Then:
$$e_1(e_2e_4) = e_1e_6.$$
From the line (e_1, e_7, e_6), the directed order is $e_6e_1 = e_7$, so $e_1e_6 = -e_7$. Thus $e_1(e_2e_4) = -e_7$.

Comparing the two evaluations:

$$(e_1e_2)e_4 = e_7 \neq -e_7 = e_1(e_2e_4).$$

In fact, $(e_1e_2)e_4 = -e_1(e_2e_4)$, which proves that the octonion algebra $\mathbb{O}$ is non-associative. $\square$